# To Thine Own $\sigma$ Be False:

## A network's learned $\sigma$-profile compensates for a misspecified closure rather than reporting the molecule's charge distribution

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**Abstract**

Grounding a learned physical latent in reference values that a black-box model cannot exploit is a common recipe for out-of-distribution prediction, on the expectation that more accurate physical inputs yield more accurate predictions. We show that this expectation fails when the grounded inputs are consumed by a fixed, misspecified thermodynamic model. A graph neural network predicts solid–liquid solubility more accurately from its own learned molecular charge-density profiles than from external quantum-chemical reference profiles fed to the same fixed activity-coefficient model (COSMO-SAC). Because the learned latent has an external reference, the error incurred by using reference inputs can be separated, without fitting any model, into a term due to misspecification of the fixed model and a term due to insufficiency of the learned inputs; the misspecification term appears to be the larger, so the fixed model, not the learned inputs, sets the accuracy ceiling. Trained end to end through the misspecified model, the latent departs from the physical quantity it nominally represents and becomes a low-rank, transferable surrogate that compensates for the model's error. As a result, more accurate physical inputs expose the model's bias instead of improving prediction, and increasing the model's fidelity does not remove the penalty. The same behaviour arises in a controlled synthetic family of models and in a second chemical domain, the prediction of acid dissociation constants. We introduce a general, fitting-free instrument that separates model misspecification from input insufficiency for any predictor that composes a fixed physical model over a learned intermediate, and we show that the interpretability such physically grounded latents appear to offer is not guaranteed when the fixed model is misspecified.

# 1 Introduction

The solubility of a solid in a liquid solvent decides how a drug is formulated, how a product is crystallised and purified, and which synthetic routes are practical, yet it remains hard to predict from molecular structure alone. Measurements of the same system disagree substantially between laboratories, and the property depends jointly on how a molecule interacts with the solvent and on the energetics of its crystalline solid, so first-principles calculation is costly and any data-driven predictor must generalise to molecules unlike those it was trained on.

Two broad strategies have emerged. Purely data-driven models regress solubility directly from molecular descriptors or learned representations; they are at present the most accurate on unseen molecules, but they are opaque and offer no account of why a given prediction succeeds or fails. Physics-informed models instead route the prediction through interpretable thermodynamic quantities—an activity coefficient that describes the solute–solvent interaction, and the melting properties of the crystalline solid—each of which can be anchored to abundant single-component reference data that a black-box regressor cannot exploit. The appeal of this second strategy rests on a widely held expectation: that supplying a model with physical inputs closer to their true values should make its predictions more accurate.

We find that this expectation can fail, and we explain why. We study a physics-informed solubility model whose learned intermediate is a molecular charge-density profile (a $\sigma$-profile) that is consumed by a fixed thermodynamic model, COSMO-SAC. Replacing the model's own learned profiles with external quantum-chemical reference profiles makes its predictions worse rather than better. The reason is that the fixed model is misspecified, and the learned profile, trained end to end through it, has reorganised itself to cancel the model's error rather than to report the physical quantity it nominally represents. Supplying a more faithful profile removes this compensation and exposes the model's underlying bias.

Diagnosing such a failure is difficult because the error of a composed predictor confounds two causes: the fixed model may be wrong, or the learned inputs may carry too little information. Our central methodological result is that when the learned intermediate has an external reference, these two causes separate exactly, without fitting any further model, and the separation attributes the accuracy ceiling to the fixed model rather than to the inputs. We then show that the compensating behaviour of the learned intermediate is systematic and transferable, that raising the fidelity of the fixed model does not lift the ceiling, and that the same phenomenon arises in a controlled synthetic setting and in a second chemical domain. Together these results qualify a common assumption of physics-informed modelling: grounding a learned latent on reference values helps only when the model that consumes it is faithful, and the interpretability such a latent appears to offer is not guaranteed when it is not.

**Contributions.** This article makes the following contributions.

- We report a grounding paradox in physics-informed solubility prediction: a graph neural network predicts solubility more accurately from its own learned $\sigma$-profiles than from external quantum-chemical reference profiles supplied to the same fixed COSMO-SAC model, reversing the expected benefit of more accurate physical inputs.

- We introduce a fitting-free instrument that uses an external reference for the learned latent to split the prediction error into a model-misspecification term and an input-insufficiency term, and we use it to show that the misspecification of the fixed model is the larger term and sets the accuracy ceiling.

- We identify the mechanism behind the paradox: trained end to end through the misspecified model, the learned latent becomes a low-rank, transferable surrogate that compensates for the model's error instead of reporting the physical quantity, which is why more accurate inputs hurt.

- We show that the ceiling is a property of the fixed model rather than of downstream processing: a higher-fidelity fixed model does not remove the penalty on representative data, and a post-hoc correction of the model's output restores calibration but not accuracy.

- We show that the account is general. It applies to any predictor that composes a fixed physical model over a learned intermediate, and the same behaviour appears in a controlled synthetic family of models and in a second chemical domain, the prediction of acid dissociation constants.

The remainder of this article is organised as follows. Section 2 reviews data-driven and physics-informed approaches to solubility prediction and the broader question of when imposed physical structure helps or hurts a learned model. Section 3 develops the theory for a composed predictor and defines the exact separation of model misspecification from input insufficiency. Section 4 describes the model, the data, and the reference-input substitution. Section 5 presents the solubility paradox, the measurement that attributes the accuracy ceiling to the fixed model, the compensating-surrogate mechanism, and the synthetic and second-domain replications. Section 6 discusses the implications and the limitations, and Section 7 concludes.

## 2 Related work

Two families of methods dominate the prediction of solid solubility and solvation, and they make opposite choices about physical structure. Physics-informed pipelines such as SolProp[2] assemble a thermodynamic cycle from separately learned components (solvation free energy, vapour pressure, fusion properties), so that each prediction is decomposable and its intermediate quantities carry physical meaning. Direct models forgo that structure: FastSolv[3] regresses solubility from molecular descriptors with a deep neural network and is the current leader on extrapolation to new solutes, a task on which inter-laboratory disagreement sets a substantial irreducible-error floor. On accuracy the ordering is clear: the black box leads and the physics-informed cycle trails it.

The activity coefficient is the quantity a solubility model must supply, and learned surrogates for it have advanced quickly. Winter et al.[4] predict infinite-dilution activity coefficients $\ln \gamma_2^\infty$ directly from SMILES with a language model, and Sanchez-Medina et al.[5] embed a Gibbs–Helmholtz constraint in a graph network to capture their temperature dependence. These approaches treat the activity coefficient as the regression target, rather than retaining a mechanistic model that consumes a predicted molecular property.

The broader question of when a fixed physical structure helps or hurts a learned model is central to physics-informed machine learning[6]. Its dominant forms impose structure as a differentiable object: physics-informed neural networks enforce a governing equation through the loss[7], while physics-guided and gray-box hybrids compose learned components with fixed mechanistic maps[8], and a recent operator-learning approach adds a trainable correction to a misspecified governing equation[9]. The prevailing expectation is that more physics improves generalisation, yet the same structure can degrade it: physics-informed networks have documented failure modes in which the imposed constraint makes optimisation or accuracy worse[10]. In neural solvers of partial differential equations, Mohan et al.[11] show that a learnable term added to a fixed discretised operator can absorb the simulation's truncation artifacts rather than the physics, so that the program's apparent interpretability is illusory; their error source is numerical, and they have no external reference for the learned term.

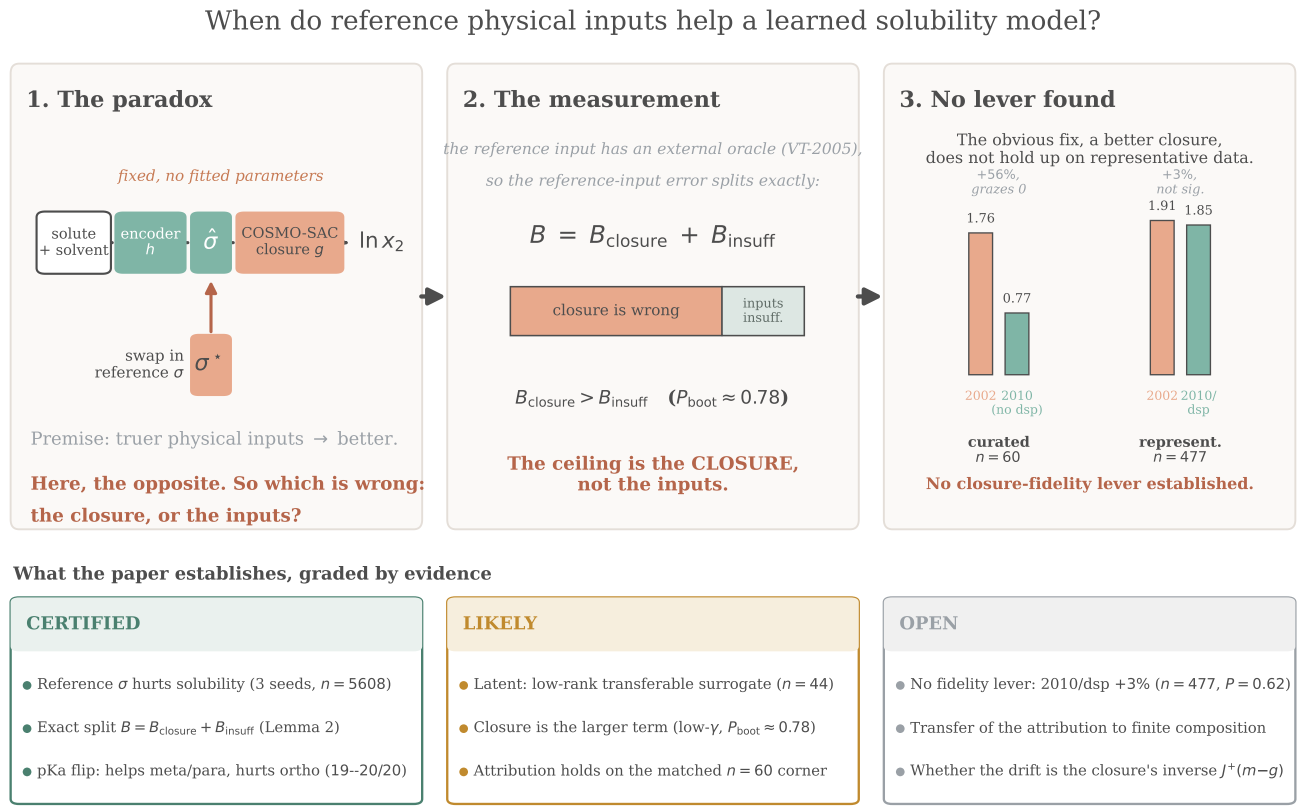


Figure 1: **Overview.** A learned encoder $h$ predicts a molecular charge-density profile that a fixed, parameter-free thermodynamic model $g$ (COSMO-SAC) turns into a solubility. *(1) The paradox:* replacing the learned profile $\hat{\sigma}$ with an external quantum-chemical reference profile $\sigma^\star$ makes the prediction worse. *(2) The measurement:* the error from using reference inputs is split into a model-misspecification term $B_{\text{closure}}$ and an input-insufficiency term $B_{\text{insuff}}$, with the misspecification term the larger. *(3) No fidelity lever:* a higher-fidelity model does not remove the penalty on representative data. The lower panel lists the paper's findings, graded by strength of evidence.

The same concern appears in concept-bottleneck models as *concept leakage*, in which a learned concept silently encodes information beyond its nominal meaning and its interpretation becomes misleading[12–14]. Recent work traces one route to leakage to the downstream model rather than to a trainable head: an inflexible or misspecified head forces the encoder to deform the concept into carrying a dependence the head cannot represent[15], and overwriting a frozen head's leaked concepts with their ground-truth values can then degrade accuracy[16].

Several diagnostic tools are relevant to such failures. The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O'Hagan[17] and its calibration critique[18]; separating whether predictions are right on average from whether they are sharp is the calibration and reliability–resolution decomposition of Murphy and DeGroot[19,20]; and a misspecified model still converges to a well-defined pseudo-true limit under the quasi-maximum-likelihood account of White[21]. That a decomposition may be only weakly constrained by the data is the subject of parameter sloppiness[22,23] and of structural versus practical identifiability[24,25], and the risk that apparent skill reflects pipeline choices rather than physics is the concern of underspecification[26], leakage and reproducibility[27], and compositional risk[28].

What this literature lacks is a way to *measure*, rather than merely diagnose, whether a fixed physical model or its learned inputs limits a composed predictor's accuracy, and to test whether the learned intermediate has become a surrogate that compensates for the model's error. The concept-leakage literature identifies the phenomenon but has no external reference against which to quantify it, and the neural-solver work has a fixed operator but only a numerical, unreferenced error term. We supply the missing ingredient: an external reference for the learned intermediate—a reference σ-profile for solubility, and tabulated substituent constants for a second chemical domain—which turns the qualitative concern into a model-free measurement that separates model misspecification from input insufficiency. Using it, we show on a fixed thermodynamic model that the learned intermediate becomes a low-rank, transferable surrogate, and that grounding it on reference values can therefore reduce accuracy. Our data are drawn from BigSolDB 2.0[29]; FastSolv and SolProp remain the accuracy reference points, which we do not contest.

# 3 Setup and theory

For a solute $s$ in solvent $v$ at temperature $T$, the target is $y = \ln x_2$, the log saturated mole fraction. Dissolving a solid trades two costs against each other—breaking the crystal lattice ($\Phi$) and mixing the freed molecules against non-ideal solute–solvent interactions ($\ln \gamma_2$)—and solid–liquid equilibrium (SLE) sets the log-solubility to their negative sum:

$$\ln x_2 = -\underbrace{\frac{\Delta H_{\text{fus}}}{R}\Big(\tfrac{1}{T} - \tfrac{1}{T_m}\Big)}_{\Phi(T)\ \text{(crystal)}} - \underbrace{\ln \gamma_2(x_2, T)}_{\text{activity}}, \tag{1}$$

with the constant-$\Delta C_p$ (heat-capacity-of-fusion) correction omitted for clarity. The activity term is produced by a *differentiable closure* $g$: a graph encoder predicts a $\sigma$-profile—the distribution of screening (surface) charge density over a molecule, the histogram a COSMO model consumes—for each molecule, and COSMO-SAC (the COnductor-like Screening Model–Segment Activity Coefficient[30]) maps the pair of profiles to $\ln \gamma_2$. Write $z^\star$ for the *reference* physical latent, the externally referenced $\sigma$-profile from the VT-2005 database (the Virginia Tech 2005 compilation of quantum-chemically computed screening-charge profiles for ~1,400 compounds[31]), and $g(z^\star)$ for the closure output when fed the reference profile. The model's own head instead predicts a *learned* profile $\hat{z}$ and evaluates $g(\hat{z})$; a matched *DirectGNN* control shares the encoder but emits $\ln x_2$ with no closure, and is the reference for "what the physics bottleneck costs" (schematic in SI, Fig. 6).

**The oracle substitution.** Feeding $z^\star$ in place of $\hat{z}$ (a $\sigma$-oracle) isolates the closure from encoder error: if the closure were well specified, $g(z^\star)$ should be the best available activity estimate. The paradox is that it is not.

**Data.** Solubility measurements are a Bemis–Murcko scaffold split of BigSolDB[29] (grouping molecules by ring-system-plus-linker scaffold, side chains stripped), so test solutes share no scaffold with training; the activity labels $m$ are experimental infinite-dilution activity coefficients ($\ln \gamma_2^\infty$, IDAC) compiled from ThermoML (the NIST/IUPAC archive of experimental thermophysical data[32]), and the reference $\sigma$-profiles $z^\star$ are the VT-2005 database.

The results below are stated for a general *composed predictor* $\hat{y} = g(h(x))$ in which a fixed, non-fitted map $g$—the *closure*—consumes a learned physical intermediate $z = h(x)$, and are compared against a matched free predictor that shares the encoder but drops $g$. Solubility, with $g$ the COSMO-SAC closure and $z$ the $\sigma$-profile, is one instance; §5.7 gives a second (pKa through a fixed Hammett closure), and §5.5 a controlled synthetic family. Nothing in Lemmas 1–2, Theorem 1, or Proposition 1 is specific to COSMO-SAC. Two quantities organize the section: the closure/insufficiency split $B = B_{\text{closure}} + B_{\text{insuff}}$, which is the assumption-light *instrument*, and the observable grounding gap $\Gamma$, which is an *indicator* of closure misspecification whose validity we characterise.

## 3.1 Why a free head can compensate: structural non-identifiability

The crystal/activity split is not identifiable from solubility data alone, which is precisely why an unconstrained head can silently absorb crystal error into the activity term.

**Lemma 1** (Structural non-identifiability of the SLE split)**.** *Consider the infinite-dilution SLE observable* $\mu(T) = -\Phi(T) - \ln \gamma_2^\infty(T)$ *with* $\Delta C_p = 0$*, so that* $\Phi(T) = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$ *is affine in* $u = 1/T$*, and a symmetric two-suffix activity class* ($\ln \gamma_2 = \ln \gamma_2^\infty (1 - x_2)^2$) *in which* $\ln \gamma_2^\infty(T)$ *is affine in* $u$*. Then, for any design with at least two distinct temperatures,* $\mu$ *depends on the parameters only through the two-dimensional statistic (its slope and intercept in* $u$*), and the activity intercept and slope already span that plane; hence the crystal parameters* $(\Delta H_{\text{fus}}, T_m)$ *are unidentified, the profiled Fisher information for* $\Delta H_{\text{fus}}$ *is exactly zero, and the indeterminacy set is two-dimensional. The deficiency is closed only by a rank-completing external constraint (a direct crystal label, or a fixed activity slope). At finite dilution the composition factor* $(1 - x_2)^2$ *breaks the exact affinity, leaving a non-zero but near-degenerate information of order* $x_2^2$ *(numerically, condition number* $2.3 \times 10^5$*; §G.2).*

This is the standard structural-identifiability picture for a collinear design[24,25]—equivalently the null-space picture of regularization theory, in which the component of an inverse fit the data leaves unconstrained lives in the null space of the forward map and is fixed only by the prior[33,34], restated for learned reconstruction as null-space networks[35]. What is specific to our setting is that this indeterminate component is not projected out by construction but is left free under *end-to-end* training of the SLE-coupled latent. We prove it in the SLE form (App. C) and audit it numerically in §G.2. It is *enabling background*, not our headline: it explains why grounding an under-determined factor on external single-component data is necessary, and why compensation between the factors is expected. By itself it says nothing about whether grounding helps. That depends on whether the closure that consumes the grounded inputs is correctly specified, which is the question we make measurable next.

The non-identifiability is not binary but graded by the activity class: the efficient information for $\Delta H_{\mathrm{fus}}$ is zero when the activity nuisance spans the crystal score's directions $\{1, 1/T\}$ and strictly positive (a finite Cramér–Rao bound) when the slope is constrained (Lemma 3, App. C). The numerical audit (§G.2) shows the honest corollary: merely restricting the activity slope leaves $\mathrm{sd}(\Delta H_{\mathrm{fus}}) \approx 8{,}251$ J/mol—finite but unusable—whereas external crystal labels, which contribute a score direction outside the activity tangent space, close it to $\approx 1{,}689$ J/mol.

## 3.2 The closure–insufficiency decomposition: the instrument

Let $m$ be the activity target (experimental $\ln \gamma_2^\infty$) and $g(z^\star)$ the closure evaluated on the true latent. The risk of the reference-input path—its mean squared error $\mathbb{E}[(m - g(z^\star))^2]$—decomposes exactly into a part the intermediate cannot resolve (the irreducible Bayes term) and a part the fixed map gets wrong; the excess over the Bayes predictor $\mathbb{E}[m \mid \cdot]$ is the second term alone.

**Lemma 2** (Exact closure–insufficiency decomposition and its one-sided bounds)**.** *For a* fixed *(non-fitted) closure $g$,*

$$\underbrace{\mathbb{E}\big[(m - g(z^\star))^2\big]}_{\textit{ref.-input MSE}} = \underbrace{\mathbb{E}\big[\mathrm{Var}(m \mid z^\star)\big]}_{B_{\mathrm{insuff}}\ (\textit{irreducible})} + \underbrace{\mathbb{E}\big[(\mathbb{E}[m \mid z^\star] - g(z^\star))^2\big]}_{B_{\mathrm{closure}}\ (\textit{decoder bias})}. \tag{2}$$

*The cross term vanishes identically because $\mathbb{E}[m \mid z^\star] - g(z^\star)$ is $z^\star$-measurable and $\mathbb{E}[(m - \mathbb{E}[m \mid z^\star]) \mid z^\star] = 0$. The split is exact with no fitting of the closure $g$—the cross term vanishes because $g$ is fixed and $z^\star$ is externally observed—so the total and $B_{\mathrm{closure}}$'s Jensen lower bound are fit-free; $B_{\mathrm{insuff}} = \mathbb{E}[\mathrm{Var}(m \mid z^\star)]$, however, must be estimated with a conditional-variance smoother whose choice affects the verdict (§5.2).*

Equation (2) is a bias–variance split of model discrepancy in the sense of Kennedy–O'Hagan[17,18]: $B_{\mathrm{closure}}$ is the systematic discrepancy of a fixed physical map, and its dominance over $B_{\mathrm{insuff}}$ is the empirical content of "the ceiling is the closure, not the inputs." We do not claim (2) as new. The affordance is that an externally grounded $z^\star$ lets us *measure* the split rather than assume it: unlike model-discrepancy calibration, which *fits* a correction from held-out observations, here $g$ is fixed and $z^\star$ externally observed, so the map's error is *attributed*, not corrected—available only where an external latent oracle exists (LFERs, COSMO / group-contribution closures, thermodynamic cycles), not in gray-box or PINN hybrids. **Exactness is contingent on $g$ being fixed**: for a *fitted* decoder the cross term does not vanish and a plug-in point split is invalid (a spurious $-0.36$ cross term appears when $\mathbb{E}[m \mid z^\star]$ is replaced by a fitted estimate). We therefore report *one-sided bounds*, not a point split (each proved in App. C):

- $B_{\mathrm{closure}} \geq (\mathbb{E}[m] - \mathbb{E}[g])^2$ by Jensen: a constant offset between the decoder's mean output and the target is misspecification, not input insufficiency, and this bound needs *no smoothness assumption.* Its one requirement is that $m$ and $g$ share the $\ln \gamma_2^\infty$ reference state, which we check directly (§5.2).
- $B_{\mathrm{insuff}} \leq \mathbb{E}[\mathrm{Var}(m \mid \mathrm{bin}(g(z^\star)))]$, a valid *upper* bound (law of total variance, since a binning of $g(z^\star)$ coarsens $z^\star$).

The Jensen constant-offset bound is convention-specific—it separates under the full combinatorial convention but inverts under the deployed residual-only default—so the headline separation in §5.2 rests on the leakage-immune law-of-total-variance bound, and the Jensen numbers are reported in App. D. Because $B_{\mathrm{insuff}} = \mathbb{E}[\mathrm{Var}(m \mid z^\star)]$ depends only on $(m, z^\star)$, it is *independent of the combinatorial convention* (full vs. residual) at fixed $z^\star$: disagreement between closures that share one reference profile is decoder disagreement and lands in $B_{\mathrm{closure}}$. A kernel that carries its own $\sigma$-averaging, by contrast, changes $z^\star$ itself, so the 2002-vs-2010 contrast (§5.2) is not a fixed-input decoder difference.

**The instrument's resolution is** $\dim(z^\star)/n$**.** Whether the split can be *point*-identified or only bounded is set by the latent dimension relative to sample size. $B_{\mathrm{insuff}} = \mathbb{E}[\mathrm{Var}(m \mid z^\star)]$ is a conditional variance in $\dim(z^\star)$ dimensions, and estimating it from $n$ observations is a nonparametric problem governed by $\dim(z^\star)/n$. On the solubility regime $z^\star$ is a 102-bin $\sigma$-profile with $n$=60 matched pairs ($\dim(z^\star)/n \approx 1.7$): the conditional variance is unestimable, so every $B_{\mathrm{insuff}}$ entry of Table 1 is a one-sided *bound* and $B_{\mathrm{closure}}$ is only lower-bounded—the one-sidedness above is a consequence of this ratio, not a modelling choice. Where the latent is low-dimensional the same instrument point-identifies: the pKa Hammett constant is a *scalar* with $n$=1004 ($\dim(z^\star)/n \approx 10^{-3}$), so $B_{\mathrm{insuff}}$ is a point estimate (§5.7), not a bound. The decomposition is identifiable exactly where $\dim(z^\star)/n$ is small—which is why the pKa domain, not the solubility case, is where the split is pinned rather than bounded.

## 3.3 What the instrument resolves, and what it does not

The decomposition (§3.2) is the load-bearing measurement—assumption-light, needing neither a representable encoder nor a lossless bottleneck. Alongside it we report a cheaper, purely observable quantity: the *reference-input penalty* $\Gamma := R_{\text{orc}} - R_{\text{e2e}}$, the oracle risk minus the best end-to-end risk; the paradox is $\Gamma > 0$. Under a representable encoder (A1) and a lossless bottleneck (A2) it is sandwiched $0 \le \Gamma \le B_{\text{closure}}$, so an observed $\Gamma > 0$ would certify $B_{\text{closure}} > 0$ (Theorem 1, App. C). But A1 is empirically doubtful (the encoder is transfer-limited: a linear probe drops from train $R^2 \approx 0.76$ to test $R^2 \approx 0.45$) and A2 is unverifiable and physically dubious; when A2 fails, $\Gamma > 0$ conflates closure misspecification with input insufficiency—a well-specified closure with a lossy latent gives $B_{\text{closure}} = 0$ yet $\Gamma > 0$ (§5.5, and the pKa ortho sweep of §5.7). We therefore read $\Gamma$ as corroborating *evidence* and base the model-dominance conclusion on the decomposition, which needs neither assumption: *the decomposition is the instrument; $\Gamma$ is the inexpensive indicator that motivates it.*

**When a fixed prior loses; and scope.** The deployed question—does the *trained* physics model beat the *trained* black box at sample size $n$?—follows a sign rule. Writing each trained risk as an asymptote plus estimation variance, $\mathbb{E}[\hat{R}_{\text{phys}}(n)] \approx R_{\text{e2e}} + V_{\text{phys}}(n)$ and $\mathbb{E}[\hat{R}_{\text{direct}}(n)] \approx R_{\text{direct},\infty} + V_{\text{direct}}(n)$ with $V(n) \downarrow 0$, the *asymptotic physics penalty* is $\mathcal{T}(n) \approx \Delta_\infty + [V_{\text{phys}}(n) - V_{\text{direct}}(n)]$ with $\Delta_\infty := R_{\text{e2e}} - R_{\text{direct},\infty}$.

**Proposition 1** (Sign of the asymptotic physics penalty). *(i) $\mathcal{T}(n) < 0$ (physics helps) iff $V_{\text{direct}}(n) - V_{\text{phys}}(n) > \Delta_\infty$: the variance a constrained prior saves must exceed its asymptotic bias. (ii) If $\Delta_\infty > 0$ and the variance gap vanishes ($V_{\text{direct}} - V_{\text{phys}} \to 0$), there is a critical budget $n^\star$ beyond which $\mathcal{T}(n) > 0$ for all $n > n^\star$. (iii) Since $V_{\text{direct}} - V_{\text{phys}} \le V_{\text{direct}}$, as the black box nears the aleatoric floor ($V_{\text{direct}} \to 0$) the penalty tends to $\Delta_\infty$; this limit is $\ge 0$ when the control is asymptotically Bayes ($R_{\text{direct},\infty} = R_{\text{free}}$), not merely from $V_{\text{direct}} \to 0$.*

The consequence for grounding is (ii): a fixed-closure prior with strictly positive asymptotic bias always loses once the budget passes $n^\star$ (proof in App. C). There is no "optimal grounding" theorem, and the decomposition does not improve out-of-distribution accuracy (§5). Everything here is measured at infinite dilution, 298 K, on the VT-2005-matchable low-$\gamma$ regime; transfer to the finite-composition regime the paradox lives in is a conjecture, not a result (§5.8). We read these results against known work on model discrepancy[17,18], parameter sloppiness[22,23], identifiability[24,25], misspecification[21], and underspecification[26].

# 4 Methods

The model is a controlled comparison: a shared graph backbone feeds either an explicit thermodynamic bottleneck (*TGNNSolv*) or a direct regressor (*DirectGNN*) that shares the backbone but emits $\ln x_2$ with no physics. The gap between them measures the cost or benefit of the bottleneck. This section describes the shared backbone, the three interchangeable activity closures, the solid–liquid-equilibrium (SLE) solver and its implicit differentiation, the oracle substitution, and the training protocol. Exact architecture and optimisation hyperparameters and the load-bearing loss weights are in the SI (§B); the full feature list and the complete ∼30-term loss taxonomy are released with the code.

## 4.1 Data, splits, and labels

Solubility measurements are drawn from BigSolDB 2.0[29] and partitioned by Bemis–Murcko scaffold so that every test solute is unseen at training; row membership is set by a melting-point-independent miscibility/structure filter, and crystal $T_m$, $\Delta H_{\text{fus}}$, Hansen parameters, and infinite-dilution activity coefficients $\ln \gamma_2^\infty$ are merged on as per-solute labels where available. Two data hazards are handled explicitly. Melting-point tables arrive in either °C or Kelvin and are auto-detected by column median (a past double-conversion inflated every $T_m$ by 273 K, which propagates directly into the ideal term); and the seeded scaffold split is not stable across pipeline versions, so every regeneration is treated as a new split with all downstream artifacts recomputed. Crystal supervision is scarce: the test and validation splits contain essentially no measured $\Delta H_{\text{fus}}$, which is the quantitative backbone of the identifiability argument (§3). External single-component data used for grounding are an open crystal-property pool (∼15,000 melting points), the VT-2005 $\sigma$-profile database, and ThermoML infinite-dilution activity coefficients; each is filtered to exclude any molecule whose scaffold appears in the validation or test split.

## 4.2 Graph encoder and pair representation

Solute and solvent are encoded by a *shared* message-passing graph network; two other interchangeable backbones (a graph-transformer and a physics-channelled variant that supervises message channels toward the Hansen parameters) are described in the SI (§B) and do not outperform the headline MPNN. Small hydride solvents such as water are given explicit hydrogens so the polar and hydrogen-bond channels activate. Solute and solvent embeddings interact through a bipartite cross-attention/message-passing block and a physics-aware readout, and are combined into a pair representation $[g_s, g_v, g_s \odot g_v, |g_s - g_v|]$ passed through an MLP, optionally augmented with RDKit descriptors, Morgan fingerprints, and an ionic-context feature. All heads below consume this pair representation; the DirectGNN control consumes the identical representation.

## 4.3 Crystal head and the ideal term

A fusion head predicts the two crystal properties defining the ideal (activity-free) solubility, with physically bounded parametrisations ($T_m \in [100, 700]\,\mathrm{K}$, $\Delta H_{\mathrm{fus}} > 0$; an optional Walden-constrained entropy mode and a group-contribution $T_m$ prior are described in the SI). They give the ideal term

$$\Phi(T) = \frac{\Delta H_{\mathrm{fus}}}{R}\left(\frac{1}{T} - \frac{1}{T_m}\right), \tag{3}$$

the constant-$\Delta C_p$ correction omitted (audited in §F).

## 4.4 Three interchangeable activity closures

The activity term $\ln\gamma_2$ is produced by one of three closures, selected by configuration, all differentiable and all consuming the same pair representation.

**Two baseline closures.** NRTL (Non-Random Two-Liquid[36]), the most expressive of the three, predicts two interaction energies with a $1/T$ temperature law and a non-randomness factor, optionally regularised by a UNIFAC group-contribution prior[37]; both are standard and serve mainly as baselines. Its symmetric one-parameter collapse `gamma_inf` gives the analytic infinite-dilution law $\ln\gamma_2(x_2, T) = \ln\gamma_2^{\infty}(T)\,(1 - x_2)^2$ with $\ln\gamma_2^{\infty}$ affine in $1/T$, the minimal model in which the identifiability analysis is cleanest (§3).

**Differentiable COSMO-SAC.** A $\sigma$-profile head emits, for each molecule, a 51-bin screening-charge-density profile $p(\sigma)$ (area per bin, summing to the cavity area $A$). A parameter-free COSMO-SAC layer (Lin–Sandler / VT-2005 conventions) maps the pair of profiles to $\ln\gamma_2$ by a damped fixed point for the segment activity coefficient $\Gamma_S$ (distinct from the grounding gap $\Gamma$),

$$\Gamma_S(\sigma_m) = \Big[\textstyle\sum_n p_{\mathrm{norm}}(\sigma_n)\,\Gamma_S(\sigma_n)\,\exp\big(-\Delta W(\sigma_m, \sigma_n)/RT\big)\Big]^{-1}, \tag{4}$$

with a precomputed exchange-energy matrix $\Delta W$ (electrostatic misfit plus a hydrogen-bonding term); the restoring contribution is $\ln\gamma_2^{\mathrm{res}} = (A_2/a_{\mathrm{eff}})\sum_m p_2^{\mathrm{pure}}(\sigma_m)[\ln\Gamma_{S,\mathrm{mix}}(\sigma_m) - \ln\Gamma_{S,2}^{\mathrm{pure}}(\sigma_m)]$, and an optional Staverman–Guggenheim combinatorial term uses the molecular cavity volume and area. The combinatorial size factor $r_i = V_i/r_0$ takes the *reference* COSMO cavity volume ($V_{\mathrm{cosmo}}$) so that it shares the cavity basis of the area $A_i$. This closure — COSMO-SAC evaluated over a network- *predicted* $\sigma$-profile, and the intervention that swaps that profile for the external reference one — is the object of study in the grounding analysis (§5.1).

**In-house closures are the primary instrument.** Both closures used in the fidelity measurement (§5.2) are our own differentiable implementations, so that every arm shares one code path, removing the cross-implementation engine confound. Each arm consumes its own model-prescribed $\sigma$-averaging (Mullins for 2002, Hsieh for 2010); this is the specified use of each closure, and the family admits no input-fixed contrast (§5.2). The 2002 layer above reproduces the NIST reference COSMO-SAC-2002[38] to RMSE 0.003 $\ln\gamma$ on identical VT-2005 profiles. For the modern arm we implement a differentiable COSMO-SAC-2010/dsp layer[39,40] — the donor/acceptor-typed three-profile representation, the temperature-dependent electrostatic misfit $c_{\mathrm{ES}}(T) = A_{\mathrm{ES}} + B_{\mathrm{ES}}/T^2$, the typed hydrogen-bonding kernel, and the optional dispersion term — which reproduces the NIST COSMO-SAC-2010 reference to RMSE $1.4 \times 10^{-4}$ $\ln\gamma$ (dispersion term and temperature handling to RMSE 0 and $\sim 10^{-5}$). We report the fidelity contrast on these in-house layers and cross-check each point against the NIST `cCOSMO` reference.

## 4.5 The SLE solver and implicit differentiation

Given $\Phi(T)$ and a closure for $\ln\gamma_2$, the saturated mole fraction solves $\ln x_2 = -\Phi(T) - \ln\gamma_2(x_2, T)$, a fixed point in $x_2$ because $\ln\gamma_2$ depends on composition. We iterate a damped map

$$x_2^{(k+1)} = \lambda\,\exp\big(-\Phi(T) - \ln\gamma_2(x_2^{(k)}, T)\big) + (1-\lambda)\,x_2^{(k)}, \tag{5}$$

with closure-specific kernels for the NRTL, `gamma_inf`, and COSMO-SAC paths. Rather than backpropagating through the iterations, gradients use the implicit-function theorem at the converged $x_2^{\star}$ (with $\eta = \partial\ln\gamma_2/\partial x_2$): exact at the fixed point, memory-constant in the iteration count, and well conditioned unless $1 + x_2^{\star}\eta \to 0$, an explicit stability criterion. Iteration counts (train 16, eval 30) were set by a convergence audit — a shorter $n$=8 schedule failed a 5.86-$\ln x_2$ convergence test on strongly non-ideal pairs (SI §B).

## 4.6 Adaptive correction and the oracle substitution

A bounded, gated "adaptive physics correction" perturbs the physical parameters ($T_m, \Delta H_{\mathrm{fus}}, \tau$) rather than the output and re-solves the SLE, keeping the correction in parameter space to preserve the physical decomposition. The *oracle substitution* replaces the predicted $\sigma$-profile $\hat{z}$ with the reference VT-2005 profile $z^{\star}$ before the closure (matched by canonical SMILES); it is applied three ways to rule out wiring artifacts—at evaluation only, injected at training time so the model co-adapts, and as a channel-swap control—and an analogous override substitutes reference crystal targets. The activity target for the decomposition (§5.2) is the infinite-dilution $\ln\gamma_2^\infty = \lim_{x_2\to 0}\ln\gamma_2$ at 298 K.

### 4.7 Training: three-phase curriculum and single-component grounding

Optimisation follows a fixed three-phase curriculum: (P1) auxiliary-property pretraining on crystal, Hansen, and $\ln\gamma_2^\infty$ targets with no SLE loss; (P2) full SLE training, with the adaptive correction unfrozen partway and an optional crystal→pair weight ramp; (P3) low-learning-rate stabilisation with monotonicity and van't Hoff regularisers. External single-component data enter through *grounding streams*: each stream is a sidecar of *self-solvent* rows (solvent = solute, no solubility label, only the target factor's mask active), interleaved into P1/P2 at a configurable step budget, with a mandatory scaffold-leak guard excluding any pool molecule whose scaffold is in validation or test. The reference streams ground the crystal factor (melting-point pool) and the activity factor ($\sigma$-profile database); this single-component-grounding pattern is the mechanism by which an otherwise non-identifiable factor (§3) is pinned by abundant external data a black-box predictor cannot use.

### 4.8 Objective

The training loss is a weighted sum with per-phase weights. Only a small set of terms is load-bearing for the claims in this paper: a bin-weighted Huber on $\ln x_2$; auxiliary losses on the $\sigma$-profile (an earth-mover distance), on crystal $T_m/\Delta H_{\text{fus}}$, on $\ln\gamma_2^\infty$, and on Hansen parameters; and structural regularisers for temperature monotonicity and the van't Hoff slope. The remaining terms (channel orthogonality, contrastive, mixture-of-experts balance, and various priors) are auxiliary; the complete taxonomy and weights are tabulated in the SI (§B).

### 4.9 The DirectGNN control

DirectGNN shares the encoder, interaction, readout, and pair representation, adds a thermometer temperature encoding, and maps directly to $\ln x_2$ with a bin-weighted Huber loss and no heads, closure, or solver. Because it shares everything but the thermodynamic bottleneck, the TGNNSolv−DirectGNN gap is attributable to the bottleneck rather than to representation, and DirectGNN is the reference for every asymptotic physics penalty comparison in §5.

### 4.10 Reproducibility

All controlled comparisons are run at three seeds and reported as mean±sd on a cross-arm intersection lock (rows supervised and finite in every arm). Solver iteration counts, COSMO-SAC constants, architecture and optimisation hyperparameters, the load-bearing loss weights, and the compute ledger (which results are full-corpus GPU runs versus CPU-reproducible proxies) are given in the SI (§B, §I); the released code regenerates every figure and table from committed artifacts.

## 5 Results

The mechanism (§5.4) and the instrument that measures it (§5.2) carry the paper. Held against its reference on matched pairs, the end-to-end latent is a compensating surrogate that cancels the closure's error rather than reporting the physical quantity, and the decomposition locates the ceiling in the low-fidelity closure rather than its inputs. The grounding paradox (§5.1) is the observable effect that makes the failure visible, but does not by itself carry the attribution; whether a standard higher-fidelity closure (COSMO-SAC-2010/dsp) moves the ceiling we could not establish on representative data (§5.2). The remaining sections—the synthetic sweep (§5.5), the output-residual control (§5.6), and the pKa closure (§5.7)—stress-test and generalize the account, with further diagnostics in the ESI.

Every claim in this paper is graded by strength of evidence—certified (holds under any closure configuration, exactly proved, or bootstrap-certified with $P\to 1$ across resamples), likely (measured but sub-threshold or one-sidedly bounded on the matched $n$=60/$n$=44 sets), or open (unestablished, directional, or conjectural)—in the ledger at the foot of Fig. 1. We make no accuracy claim.

### 5.1 The grounding paradox

Substituting the external reference $\sigma$-profile for the model's learned one degrades its *solubility* prediction, and it fails the way a *misspecified map* does—not the way missing information would. The effect erases any positive $R^2$ (Fig. 2): the model predicts solubility better *from its own learned $\sigma$-profile* than when *given the external reference one* (MAE 1.85 vs 2.25, in $\ln x_2$ units), and it survives three controls—the evaluation-time oracle, injection of the reference profile *at training* (MAE 1.98), and a channel-swap (2.08), all worse than the learned-$\sigma$ arm—ruling out a wiring artifact.[1] That the fault is a wrong *map* rather than a missing input is indicated (likely; §5.2) on the activity axis, where the reference input is clean: restoring the physically *more* complete Staverman–Guggenheim combinatorial term (the entropic contribution from solute–solvent size and shape

[1]The three oracle-application controls (evaluation-only, training-time injection, channel-swap; §4) are computed on the same three-seed cross-arm lock as Table 4; their per-arm predictions are released with the code.

mismatch) *increases* error, and on $n$=60 matched pairs the fixed reference-input closure sits about 2.4× above the leakage-immune irreducible-variance floor (the load-bearing LOTV bound $B_{\mathrm{insuff}} \leq 0.62$; a more flexible regressor reaches a smaller, least-conservative floor near 0.40, §5.2), under-predicting $\ln \gamma_2^\infty$ with a systematic bias (SI, Fig. 7).

**Why the solubility paradox does not by itself carry the attribution.** The solubility effect is real and certified (three seeds, three controls), but it cannot carry the closure-versus-input attribution, for a data reason. Drug-like solutes are essentially absent from the VT-2005 reference set, so only a handful of test solutes have a reference $\sigma$-profile at all; on solubility the reference substitution therefore falls almost entirely on the *solvent* channel, and the both-channel substitution the attribution would need is available for too few solutes to support a per-molecule conclusion (a fresh $2 \times 2$ channel-substitution test lands on ≈7 solutes, too few to separate the cause either way). Computing reference profiles for drug-like solutes is possible in principle but costs hundreds of quantum-chemistry runs on large, multi-conformer molecules, for an effect already certified as a phenomenon—not worth it. We therefore carry the causal claim on the activity axis, where the reference input is clean and matched: the decomposition ($n$=60, §5.2) and the compensating surrogate ($n$=44, §5.4).

## 5.2 The closure-fidelity axis: locating the ceiling, and moving it

The cause is cleanest in the form that needs no bounds and no small-set variance estimation. The externally-referenced decomposition of §3.2 acts as an *instrument* on the $n$=60 matched regime: it asks whether a fixed closure's ceiling sits in the map or in its inputs. We demonstrate it on the $n$=60 matchable activity pairs (298 K, infinite dilution)—a regime where $\dim(z^\star)/n \approx 102/60$ leaves the split behind one-sided bounds (§3.2), read as a bounded demonstration, not a certificate. Its point-identified counterpart is the scalar-latent pKa domain (§5.7), where the same instrument resolves the split and recovers the a-priori channel assignment (there, insufficiency)—evidence the instrument works, not a reconfirmation of this verdict. Here $z^\star$ is the reference concatenated solute/solvent $\sigma$-profiles and $m$ the experimental $\ln \gamma_2^\infty$; Table 1 reports the bounds. In this regime too the ceiling sits in the *low-fidelity* closure rather than its inputs. The set spans 41 solutes and 17 solvents in 102 profile dimensions, with two formamide pairs dropped on a canonical-SMILES tautomer mismatch.

We separate the load-bearing evidence—a bound that holds under *any* COSMO-SAC configuration—from corroborating checks that are robust across a range of conventions but not strictly independent of the model fit.

**The load-bearing separation (leakage-immune).** On the deployed residual-only convention, the leakage-immune law-of-total-variance bound gives $B_{\mathrm{closure}} \gtrsim 0.85$ against $B_{\mathrm{insuff}} \lesssim 0.62$ (Table 1). This is the claim we lean on, and it is conservative by construction: the bound subtracts a *population upper* estimate of $B_{\mathrm{insuff}}$, understating $B_{\mathrm{closure}}$, so a surviving margin survives a fortiori. The finite-sample biases run the other way and are controlled—the deflated $k$-NN difference is kept out of the bound, and the LOTV binning's per-bin sparsity bias is cross-checked against the sparsity-immune RF and ridge out-of-fold estimators (§D).

The separation is stable but *not certified*. It survives leave-one-solvent, -solute, and -pair folds and deletion of the eight highest-error pairs, but a two-way cluster bootstrap places $B_{\mathrm{closure}} > B_{\mathrm{insuff}}$ in only 78% of resamples ($P_{\mathrm{boot}}$, a resampling frequency, not a posterior probability of the inequality; 90% margin spanning zero), and one deliberately aggressive out-of-fold ridge smoother inverts it ($B_{\mathrm{insuff}}$=0.88 > $B_{\mathrm{closure}}$=0.59). So "the larger" is estimator-conditional and likely, not certain; with 17/41 clusters the band is coarse (mid-0.7s to high-0.8s). One threat we can bound: label noise is unestimable here and assumed zero, but the bounds are invariant to zero-mean noise and the matched IDAC is essentially single-method (158/165 gas chromatography), so a within-method systematic large enough to close the $\approx 0.23\,(\ln \gamma)^2$ margin is implausible. The convention, leakage, $\gamma$-stratification, and leverage stress-tests are in App. D.

**The closure defect is low-rank and correctable.** That $B_{\mathrm{closure}}$ dominates locates the ceiling in the map; it does not say the map is *fixably* wrong. It is. A small structure-preserving residual on the fixed COSMO-SAC exchange-energy kernel ($\Delta = B\,\mathrm{diag}(a)\,B^\top$ over the $\sigma$-grid, symmetric, zero-initialised), fit on the *reference* VT-2005 profiles, removes 49% of the total reference-input error out-of-sample (five-fold cross-validated, activity MSE 1.47 → 0.75—most of the reducible closure component above the $B_{\mathrm{insuff}}$ floor) at *rank one* (52 parameters; rank 2 adds nothing, rank 3 over-fits). At 52 parameters against $n$=60 this is provisional, and we have not separated the kernel correction from a per-solvent offset. So $B_{\mathrm{closure}}$ is not diffuse model error but a low-dimensional, correctable defect of the map—addressable with a handful of parameters on the closure, whereas the same error is only made worse by feeding the inputs truthfully (§5.1) and untouched by an output residual (§5.6); whether an end-to-end model realises the gain is a compute-gated question we leave open.

**Where the closure fails, chemically.** The residual error is not an implementation artifact: our differentiable layer reproduces the NIST benchmark COSMO-SAC-2002[38] to RMSE 0.003 $\ln \gamma$ on identical VT-2005 profiles, with self-solvation $\ln \gamma^\infty(i,i)$ exactly zero. Yet


Figure 2: Controlled comparison on the scaffold split (3 seeds, mean±sd, $n$=5608). DirectGNN shares the encoder but has no thermodynamic model; the physics arms trail it. Replacing the model's learned $\sigma$-profile with the external reference VT-2005 profile (the $\sigma$-oracle, salmon) gives the worst arm: by MAE the model predicts solubility more accurately from its own learned profile (1.85) than from the reference one (2.25), and the reference arm's $R^2$ falls to $\approx 0$. Per-arm values are tabulated in the SI.

Table 1: Closure–insufficiency decomposition on the reference-input activity path, reported as bounds rather than a point split, under the "full" (with Staverman–Guggenheim combinatorial term) and "res" (residual-only, deployed) conventions. All input-insufficiency ($B_{\mathrm{insuff}}$) rows are valid upper bounds; the load-bearing row is the leakage-immune law-of-total-variance bound on the model-misspecification term $B_{\mathrm{closure}}$.

| Quantity | full | res |
|---|---|---|
| $\mathrm{MSE}_{\mathrm{total}} = B_{\mathrm{closure}} + B_{\mathrm{insuff}}$ (exact) | 1.80 | 1.47 |
| $B_{\mathrm{closure}}$ lower via $\mathrm{MSE}-B^{\mathrm{up}}_{\mathrm{insuff}}$ (LOTV; *leakage-immune, load-bearing*) | 1.27 | 0.85 |
| $B_{\mathrm{insuff}}$ upper (bin $g$; law of total variance) | 0.53 | 0.62 |
| $B_{\mathrm{insuff}}$ upper (RF / Ridge out-of-fold) | 0.56/0.62 | same |
| $B_{\mathrm{insuff}}$ (kNN-in-$z^\star$, convention-indep., deflated / least conservative) | 0.40 | 0.40 |
| label noise $\sigma^2_\eta$ (IDAC, unestimable here) | 0 (assumed) | |

90% of the squared error falls on strong hydrogen-bond-donor solvents (acceptor-only near-exact), placing $B_{\mathrm{closure}}$ squarely in COSMO-SAC's documented single-parameter H-bond kernel—a known limitation, not a layer idiosyncrasy, and the specific defect the donor/acceptor-typed 2010 closure was built to fix. Convention-robustly, adding the physically *more* complete Staverman–Guggenheim combinatorial term *increases* error (1.47 → 1.80) on size-asymmetric pairs—what a misspecified map does, not what missing information would (details in App. D).

**No closure-fidelity lever.** If the ceiling is the closure, the obvious remedy is a better closure. A direct test returns a negative result.

The comparison has **no input-fixed design, even in principle**. The $\sigma$-profile averaging is part of each model's *definition* (Mullins averaging for 2002, Hsieh averaging for 2010, as specified by the reference implementation and the models' own authors[38]), so a single profile cannot be held fixed across the two kernels without violating one of them. Forcing the profile literally identical denies the 2002 kernel two of three channels, an input-insufficiency confound in precisely the term the decomposition of §3.2 exists to isolate: an "insufficiency-free by construction" contrast is incoherent here, not merely unverified. Each closure evaluated on its prescribed native profile is the specified use of the model, not a convention compromise, so the fair values 1.757 (2002) and 1.911 (2010) are the correct comparison.

The fair 2002 value depends on one implementation detail. The NIST reference implementation of COSMO-SAC-2002, handed a typed three-profile database, reads the nonpolar channel only and silently discards the donor

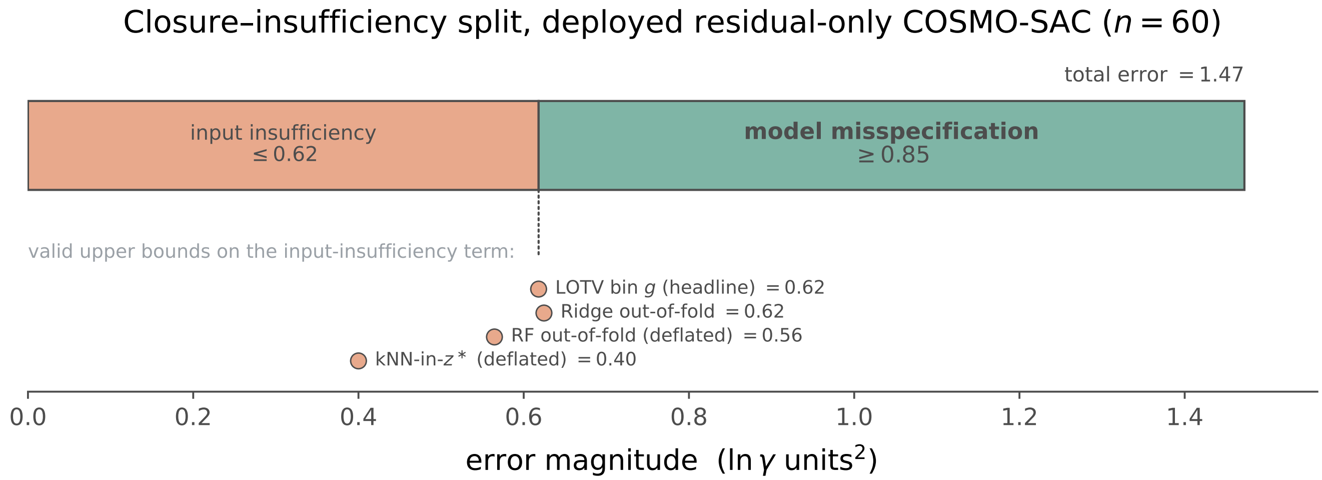


Figure 3: One-sided bounds on the closure–insufficiency split under the residual-only convention. The lower bound on the model-misspecification term ($B_{\text{closure}} \geq \text{MSE} - B_{\text{insuff}} \approx 0.85$, with MSE $= 1.47$) exceeds every valid upper bound on the input-insufficiency term $B_{\text{insuff}}$ (salmon markers); the white gap is the assumption-free separation, and the teal region is the share of the error the bound attributes to the fixed model.

and acceptor profiles while retaining the total area as prefactor; on the UD (University of Delaware) profiles this single-channel arm scores MSE 1.26, which is not the reference closure and is not comparable to the 1.47 our own layer reaches on the single-profile VT-2005 database. The fair full 2002 on the same UD data scores MSE 1.757 from our NIST-validated differentiable layer; the released `run_fidelity_lever_fair2002.py` regenerates it and cross-checks both the single-channel arm (1.26) and the 2010 arm against the cCOSMO reference.

The swap is then measurable, and it does not point the same way on curated and representative data. Across the fair 2002, the modern 2010 kernel with dispersion *off*, and 2010/dsp, the activity error is 1.757/0.765/0.785 on the curated corner and 1.911/1.978/1.855 on a representative IDAC$\cap$UD set (all temperatures, every arm validated against the NIST reference). On the corner the modern kernel cuts the error sharply ($1.757 \to 0.765$ with dispersion off, $n$=60; two-way cluster bootstrap $P = 0.98$, 90% margin $[+0.05, +2.16]$ grazing zero). On the representative set it does *not*: with dispersion off the modern kernel is *worse* than 2002 (1.978 vs 1.911), and only the dispersion term recovers a marginal +3% (1.855, $P = 0.62$, margin $[-2.25, +1.31]$). The corner reports the dispersion-off arm and the representative the dispersion arm because the transcribed dispersion term diverges from the reference on the corner's high-$\epsilon$ tail, where the dispersion value is taken from the cCOSMO reference (0.785). These are different arms of the modern family; the corner and the representative are not a single small effect.

Contribution (b), that the closure is the ceiling, is a property of the *deployed* 2002 kernel, whose fair error 1.757 sits above the insufficiency floor here and does not depend on any upgrade. On the corner the closure upgrade is in fact *better* supported ($P = 0.98$) than that closure-dominance verdict ($P_{\text{boot}} \approx 0.78$); contribution (c)—that no fidelity lever is established— therefore rests on the representative set, where the modern kernel does not beat 2002, not on the curated corner.

This is consistent with the published behaviour of the closure family rather than a challenge to it, and the comparison itself appears to be new: no large-scale infinite-dilution benchmark of the $2002 \to 2010$ step exists. The comprehensive assessment of this family[41] (29,173 $\gamma^{\infty}$ points) evaluates COSMO-SAC-2010 and COSMO-SAC-dsp only, and its largest documented infinite-dilution step—adding the entire dispersion term, $2010 \to$ dsp—buys a ~10-point mean-absolute-deviation gain ($95\% \to 86\%$), with a *reversal* on aqueous systems where dsp is worse than 2010 (203% vs. 194%). Our null for the smaller $2002 \to 2010$ step is in family with those magnitudes, and non-monotone progress across strata is documented behaviour, not an anomaly. Two things follow. COSMO-SAC-2010's demonstrated gains are concentrated in liquid–liquid equilibria and in the temperature dependence of the electrostatic term, where an infinite-dilution corner has no leverage. And the family's *best* infinite-dilution accuracy—86% MAD on 29,173 points in the hands of its own developers—is itself external evidence that the closure is the low-fidelity component: the ceiling we locate is where the field's own benchmark places it. We claim only that **we cannot establish a closure-fidelity lever on the data we can compute**, not that the modern closure is not an improvement.

The consequence for the paper is that we have no fix to offer. The ceiling we locate is not disposed of by the standard upgrade path, and the search for a remedy inside this closure family came back empty.

**The reference is itself imperfect, but conservatively so.** The external ground truth used to charge $B_{\text{closure}}$ against $B_{\text{insuff}}$ is quantum chemistry, not experiment: the VT-2005 profiles are computed for a single reference conformer and protonation state (§I); that this matters is documented rather than hypothetical—the successor Delaware database revises the VT-2005 molecular conformations for about a third of its compounds[38]. Any mismatch between that fixed state and the conformer/tautomer population under the measurement conditions enters as noise in the reference $z^{\star}$, and such noise raises $B_{\text{insuff}}$—the input-sufficiency term—not $B_{\text{closure}}$. It therefore works against the closure-dominance reading rather than for it: a truer, condition-matched reference would lower $B_{\text{insuff}}$ and widen the margin we already find, so the verdict is conservative to this defect.

**Scope.** This measurement is a low-$\gamma$/298 K/VT-2005-matchable regime and does not transfer to the finite-composition regime the paradox lives in (§5.8); the full audit is in App. D.

## 5.3 Isn't this just a bad closure?

Yes—and that is the contribution, not the objection. COSMO-SAC-2002 has a documented single-parameter hydrogen-bond kernel, and the 2010/dsp closure was built to fix exactly that defect; we do not dispute this—it is a well-documented defect, though on our data upgrading the closure did not establish a fidelity gain (§5.2). But the 2002 kernel is not an exotic choice—it is the closure a practitioner reaches for, and its fidelity is *not knowable in advance* on a new system. The field routinely composes an expressive learned encoder with a fixed mechanistic map and assumes the map is good enough. What we show is what happens when it is not: the encoder silently compensates, the resulting latent is a surrogate rather than the physical quantity it is named after (§5.4), and the interpretability the bottleneck appeared to buy is absent—none of it visible without an external oracle. Our decomposition is how you make it visible on *your* closure, before its misspecification has quietly eaten your result. The practical consequence inverts the usual instinct: before investing in richer inputs, measure the split—within a closure family the payoff is in the closure.

## 5.4 Why grounding hurts here: the end-to-end latent is a compensating surrogate

This is the *mechanism* behind the low-fidelity, end-to-end regime of the map (§6): when a $\sigma$-profile head is trained *end-to-end* through a fixed, misspecified closure—never supervised toward the reference profile—it is under pressure not to be physically faithful but to *cancel* the closure's error. The external ground truth that lets us measure the ceiling (§5.2) also lets us test this directly, comparing the predicted profile $\hat{\sigma}$ against the reference VT-2005 profile $\sigma$ molecule by molecule ($n$=44).

**The learned profile is a systematic, low-rank, transferable surrogate—confound-free.** We use the most controlled test available: a grounded model checkpointed before and after solubility fine-tuning (the $\sigma$ head unfrozen), so architecture and initialisation are held fixed and only the objective differs; we repeat it over *three seeds* and report mean±sd. The grounded base recovers the reference profile to within $36 \pm 1\%$ ($\|\hat{\sigma}_{\text{grounded}} - \sigma\|/\|\sigma\|$), so the intermediate *can* be physical; solubility fine-tuning then drives it $41 \pm 3\%$ off the grounded profile, for a total departure from truth of $51 \pm 2\%$. The drift is not per-molecule noise, and its structure is not a constant offset: the first two principal components of the *mean-centred* deviation carry $72 \pm 1\%$ of the variance, 4.8× the finite-sample isotropic null of ≈15% (the top-two sample eigenvalue share of Gaussian noise at $n$=44, $p$=51, from eigenvalue spread, not the naive 4% population value). To test whether that structure transfers across molecules we fit one distortion operator $D$, a dense affine map on the 51-bin grid (a $51 \times 51$ linear part plus a 51-bin offset, 2652 parameters). The fit is *regularised*, not free: each output bin is a ridge regression of 52 coefficients ($\ell_2$ penalty $10^{-3}$) on 22 of the 44 molecules, underdetermined without the penalty, so the held-out 22 carry the claim rather than the fit. On that held-out half $D$ predicts the drift $4.1 \pm 0.5\times$ better than the zero-drift (identity) null out-of-sample. That null is deliberately weak: the affine offset lets $D$ absorb the systematic mean shift as well, so the transfer ratio alone also credits it for that shift; the evidence that the drift has structure *beyond* an offset is the mean-centred low-rank spectrum above. So $\hat{\sigma}$ is a systematic, low-rank, *transferable* closure-compensating surrogate—one generalizable way the intermediate departs from the physical profile to compensate the closure, shared across chemistry, driven by the fitting objective rather than by architecture or a lack of grounding.[2]

[2] Three-seed mean±sd (seeds 0–2). The central claim is the *sign* and the low-rank, transferable *structure*, both seed-stable; the *magnitude* has moved across pipeline versions over the project—a short run reached only ∼16%, and earlier confounded/two-model analyses over-stated the departure (∼82%) and transfer (16–

**Dominant mode and consequence.** The dominant drift mode (Fig. 4) transfers area *out of* the nonpolar bins ($\sigma \approx 0$) and *into* the polar wings: the network makes molecules look more polar than they are, inflating the segment interactions the misspecified exchange kernel under-weights and thereby cancelling part of $B_{\text{closure}}$—the input compensating the map, exactly as the non-identifiability of the split predicts (Lemma 1). This re-explains why grounding hurts *here* (§5.1): the pipeline is calibrated to the compensated $\hat{\sigma}$, not to physics, so substituting the reference $\sigma$ breaks the compensation. It also means the model's "physical" intermediate is not the molecule's charge distribution but a surrogate, consistent with known cautions that the interpretability of physics-grounded latents is not guaranteed under misspecification[18,42], and that adapting a representation can itself collapse an otherwise physically-faithful latent[43]. Two things are specific here. The drift is the mechanism under study rather than an artifact of the evaluation: the adaptation is the solubility fine-tuning that *creates* the surrogate, and it is referenced throughout to an external profile, not read from an unconstrained probe. And its *structure*—low-rank and transferable, not diffuse—distinguishes it from a random adaptation collapse.

**Why $\sigma$ is under-determined by activity.** The freedom that makes the surrogate possible is the closure's geometry. At infinite dilution the aggregate activity Fisher $J^\top J$ ($J = \partial \ln \gamma_2^\infty / \partial \sigma$) is rank-one (participation ratio 1.0 by autograd; finite differences decorrelate the redundant flat-profile coordinates and inflate it, §F, so we read PR against a permuted control, not absolutely), and the per-molecule Fisher that governs the drift has participation ratio ≈1.4 of 51 even at full solvent coverage: the activity map concentrates on the one or two hydrogen-bonding directions where the closure is least faithful (§G.4) and is weakly sensitive to the rest, so the end-to-end head can move $\sigma$ along the low-sensitivity directions at little cost to the fit (the concrete, $\sigma$-profile form of the non-identifiability of Lemma 1). We state this narrowly—participation ratio is a concentration statistic, not a hard count, and it is measured at infinite dilution, so its transfer to the finite-composition regime that generates the drift is not established (the same scope gap as $B_{\text{closure}}$, §5.8); the external reference here is itself *computed* (VT-2005 quantum chemistry), not experimental, so this does not establish that recovery is impossible in principle.

Whether the drift is additionally the closure's *signed* first-order inverse $J^+(m-g)$ is under-resolved on this sparse, near-rank-one problem—a signed null we report but do not rely on (App. H.1); the central claim is the under-determination above, not the geometry of the drift.

26×)—so we report it but do not draw strong conclusions from it (with $n$=3 the sd is itself only indicative, not a precision).

**The contrast that fixes the sign, and how the axes interact.** The sign of grounding is not universal, and it turns on *both* axes. On our own model, holding the $\sigma$ head at its supervised warmup manifold instead of letting it drift end-to-end cuts the reference-input penalty by 65% ($\Delta$MAE from +0.40 to +0.14 over $n$=5571 matched test rows): a near-physical intermediate makes the pipeline far more robust to substituting the reference $\sigma$. But supervision alone does not *flip* the sign here—the penalty stays positive, because the closure is still the low-fidelity COSMO-SAC-2002 and the warmup $\sigma$ is itself only partly physical. The full flip needs supervision *and* fidelity together: the recent TeNNet-SAC model[1], which supervises its $\sigma$-profiles toward quantum-chemistry values *and* learns its segment-activity closure, outperforms COSMO-SAC and gains from truer profiles. So the two axes interact—supervision moves the system toward the helps-regime, closure fidelity carries it across (§6).

## 5.5 Closure fidelity sets the reference-input penalty

To isolate the mechanism we built a controlled synthetic experiment in which only the closure's fidelity varies (input quality, sample size, and estimator held fixed). A teacher draws a latent $z^\star$ and a target through a known map $T$; a fixed candidate closure $g_F$ reconstructs it under a tunable misspecification of scalar fidelity $F$ ($F = 1$ recovers $T$ exactly, smaller $F$ pushes $g_F$ away). The reference-input penalty $\Delta R^2 = R^2_{\text{oracle}} - R^2_{\text{head}}$ is $B_{\text{closure}}$ made visible—the free head recovers $\mathbb{E}[m \mid z^\star]$ while $g_F$ is the misspecified map. As $F$ falls the penalty grows monotonically and without exception: averaged over six misspecification forms (linear, quadratic, cubic, absolute-value, sinusoidal, cross-term) $F$=1.00 gives $\Delta R^2 \approx 0$, $F$=0.38 gives $\approx -0.33$, and an over-rotation stress test ($F$=−0.50) gives $\approx -2.0$ (Fig. 5); the ordering is identical across four unrelated teacher families (linear, monotone-nonlinear, kinetics-exponential, PDE-field), so the effect is a property of fidelity magnitude, not of any one domain. Placing the real system on the sweep only illustratively (the real-closure-to-$F$ map is a synthetic construction, not a calibrated measurement), the COSMO-SAC closure over predicted $\sigma$-profiles falls at $F \lesssim 0.4$: its solubility $\Delta R^2 \approx -0.36$ is what a closure of this fidelity predicts, consistent with $B_{\text{closure}}$ dominating $B_{\text{insuff}}$.

The same engine shows why the grounding gap is an indicator, not a certificate (§3.3). Holding the closure *well specified* ($B_{\text{closure}} = 0$) and varying how much target-relevant information $z^\star$ discards: when $z^\star$ is sufficient, $\Gamma$ tracks $B_{\text{closure}}$ as fidelity varies (Theorem 1's proxy is exact); when $z^\star$ is lossy but the closure is correct, $B_{\text{closure}} = 0$ throughout yet $\Gamma > 0$ grows with the discarded variance. A positive grounding gap can thus

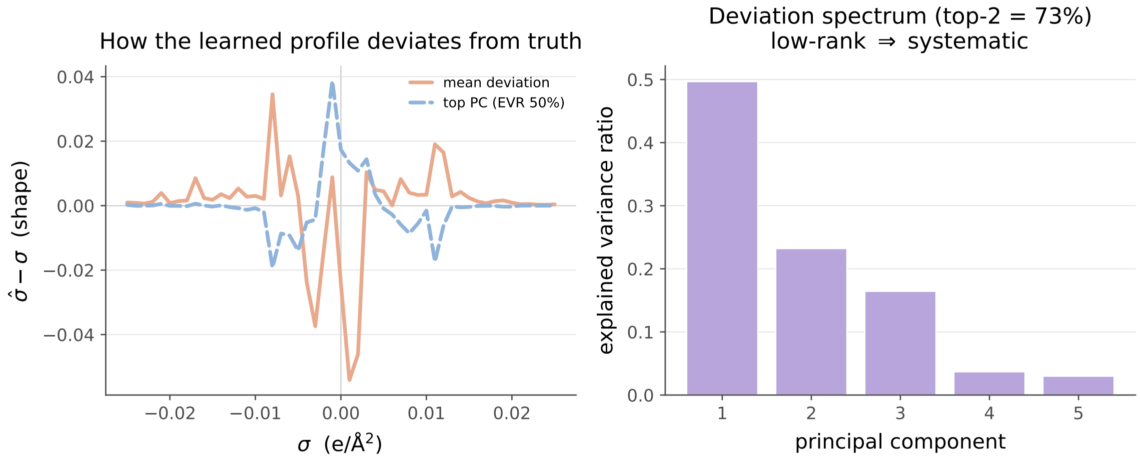


Figure 4: Deviation of the learned $\sigma$-profile $\hat{\sigma}$ from the reference VT-2005 profile $\sigma$. *Left:* the mean deviation $\hat{\sigma} - \sigma$ (and its dominant PCA mode), which transfers area out of the nonpolar bins ($\sigma \approx 0$) into the polar wings. *Right:* the deviation spectrum (top-two modes $72 \pm 1\%$, $4.8\times$ the finite-sample isotropic null of $\approx 15\%$ at $n{=}44$, $p{=}51$). A single ridge-fit affine distortion (2652 parameters, 22 train / 22 held out) transfers $4.1 \pm 0.5\times$ better than the zero-drift null out-of-sample (spectrum mean-removed); three seeds, $n{=}44$ matched VT-2005 molecules.

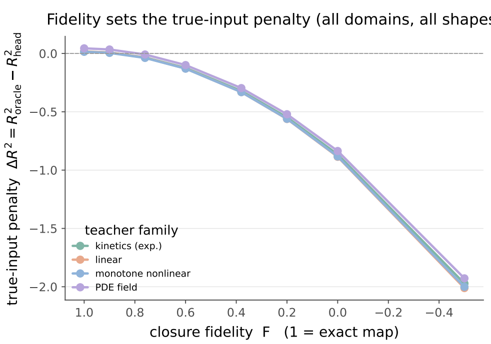


Figure 5: Reproducible closure-fidelity sweep. As the closure is made more misspecified (fidelity $F$ decreasing from 1), the reference-input penalty $\Delta R^2 = R^2_{\text{oracle}} - R^2_{\text{head}}$ grows monotonically, and the same ordering holds across four non-chemistry teacher families (linear, monotone, kinetics-exponential, PDE-field) and six misspecification functional forms. The effect is a property of the fidelity magnitude, not of any one domain or deformation. The real COSMO-SAC closure sits well inside the misspecified regime ($F \lesssim 0.4$)—the sign and rough magnitude of the empirical paradox, not a precise fidelity value.

be pure input insufficiency, not closure error—so closure-dominance is certified by the convention-independent decomposition, and $\Gamma$ only corroborates it.

## 5.6 The ceiling is not fixable downstream

If the closure binding were merely a bad point estimate, a learned output-space correction on top of the physics prediction should recover accuracy. It does not. An additive bounded residual with a learned confidence gate ($\hat{y} = \hat{y}_{\text{phys}} + (1-c)\,r$, with $y = \ln x_2$) leaves the valve effectively shut (applied residual $\approx 0$). Forcing the gate open removes the systematic *bias* ($+0.45 \to +0.06$) but recovers little *accuracy* ($R^2$ $0.236 \to 0.274$ on the same rows, a marginal rise that leaves the fit far short of usable). This is the classical calibration-versus-accuracy separation: when the upstream map is misspecified, a monotone or affine post-hoc map can fix reliability but not resolution. Recalibration buys calibration without buying accuracy, which is what we expect if the closure, rather than the estimate, is the binding constraint.

## 5.7 A second chemical closure: pKa through a Hammett LFER

The machinery is not specific to COSMO-SAC. In a second real domain—aqueous p$K_a$ of substituted aromatics—the fixed closure is a Hammett linear-free-energy relationship $g(\sigma) = \mathrm{p}K_{a,0} - \rho\,\sigma$ over the Hammett substituent constant $\sigma$ (*not* the $\sigma$-profile of §3), for which Hansch–Leo tabulated constants supply the external oracle. The two classic LFER failure modes fall cleanly onto the two channels of Lemma 2: *resonance saturation* (a curvature the linear closure cannot represent) lands in $B_{\text{closure}}$, while *ortho/steric* effects, absent from the tabulated constants, are information the intermediate lacks and land in $B_{\text{insuff}}$. On the OPERA experimental compilation[44]—the canonical full set under a uniform, closure-independent quality filter that drops pre-ionized input microstates (App. H)—this yields a *measured* fidelity sweep: the clean meta/para series ($n{=}158$) is well specified (RMSE 0.55, $B_{\text{closure}}$ below the insufficiency floor), while the ortho/heteroaromatic regime

($n$=846) degrades to RMSE 2.26, input-insufficiency dominated—the near-exact-to-degraded transition, measured rather than asserted. The decisive test is the same oracle swap as on solubility—replace the learned $\hat{\sigma}$ with the tabulated $\sigma^\star$ in the same closure and read the sign, per stratum—now certified *in-distribution* across $K$=20 within-scaffold splits with a paired per-molecule, scaffold-cluster bootstrap. It *flips*. On the clean meta/para pole feeding the reference constant *helps* (learned MAE 0.48 vs oracle 0.315; oracle better in 19/20 splits, $P$(oracle worse)<0.01): the ceiling there is the noisy learned estimate, not the closure—a real-data grounding-helps corner and an internal positive control. On the ortho pole the same swap *hurts* (learned 0.83 vs oracle 1.62; oracle worse in **20/20** splits, $P$(oracle worse)=1.00, 90% CI [+0.6, +0.9]): the freely learned $\hat{\sigma}$ absorbs the ortho/steric information the tabulated constant lacks—the compensating-surrogate mechanism of §5.4 on a second closure.

Both signs obey one *crossover law*—grounding helps a pole iff the learned arm's error there exceeds the fidelity-set oracle error—which a competence sweep confirms (App. H): the well-specified pole's oracle threshold ($\approx$0.315) is not beaten by the learned arm, so grounding helps at every training budget; the misspecified pole's threshold (1.62) is high, so the learned arm beats it, and grounding hurts, across the whole interpolation range (down to 12% of the data), reverting to helps only under scaffold *extrapolation*, where $\hat{\sigma}$ breaks (2.6±2.0) back above the threshold. The flip is thus a within-distribution, fidelity-driven crossover whose one scope boundary is predicted by the same law. It is the sharp, two-sided form of the penalty sign rule (Prop. 1); its "a fitted intermediate can beat the true one" half is a known phenomenon—from plug-in efficiency under correct specification[45] to pseudo-true fitting under misspecification[21]—that the crossover sharpens into a fidelity-set threshold. The weaker trained physics-vs-DirectGNN comparison is a complement we do not rely on (App. H). On an independent chemical closure the diagnostic separates a certified helps-corner from a certified hurts-corner. The full construction, the semi-synthetic estimator validation, the measured OPERA sweep, and the stratified comparison are in App. H (Fig. 9).

### 5.8 What we do not measure, and why

The paper's strongest claim and its causal attribution live in *different regimes*, and we state the gap plainly rather than distribute it across scope footnotes. The grounding paradox is measured where the model operates—finite composition, all $\gamma$, $n$=5608. Its *cause*—the closure/insufficiency attribution (§5.2) and the compensating-surrogate mechanism (§5.4)—is measured only on the matched external-reference regime: infinite dilution, 298 K, low $\gamma$, $n$=60 (attribution) and $n$=44 (mechanism). So the cause of the headline result is *not* measured in the headline's own regime; we transfer it by in-regime signs—the more-physics-worse effect on the full $n$=5608 set and the chemistry-resolved structure (§G.4)—which is a conjecture rather than a measured fact, and is the principal open question here. Two structural limits prevent us from closing it. The external-reference regime cannot be enlarged: only 74 VT-2005-matchable IDAC pairs exist (60 at 298 K plus 14 off-temperature), so the $n$=60 attribution cannot be pushed to significance on this data; and the matched IDAC is essentially single-method (gas chromatography), so a method-stratified re-estimate is unavailable. We therefore present the attribution as a bounded demonstration on the matched corner and mark both its regime transfer and the absence of an established closure-fidelity lever (§5.2) as the open questions.

## 6 Discussion

The result is a cautionary measurement for physics-informed modeling: grounding a model on truer physical inputs is only as good as the closure that consumes them, and when that closure is a fixed map of unknown fidelity—one the standard route does not improve—better inputs surface its bias rather than help. The external latent does not make the physics path win. Used as an instrument on the regime we can measure, the decomposition places the ceiling in the closure rather than its inputs; but that ceiling is not a lever: a fair, fully validated comparison of the 2002 and 2010/dsp kernels on representative infinite-dilution data does not establish a fidelity gain. We locate a ceiling and offer no lever. This attribution is a demonstration on a low-$\gamma$, infinite-dilution regime, not a certificate: it need not transfer unchanged to the finite-composition regime the paradox spans, where its shared closure-misspecification cause is a conjecture supported by in-regime signs (the more-physics-worse effect on the full $n$=5608 set and the chemistry-resolved structure), not a measured fact.

**What is load-bearing here.** The unit of contribution is the *instrument*, not any single measurement. Its two ingredients are individually standard: the closure–insufficiency split is a law-of-total-variance identity, and the sign of grounding's effect follows the model-discrepancy principle[17,18]—a well-specified map is helped by truer inputs, a misspecified one is not. What is new is their combination into a usable measurement. Because a learned physical latent admits an *external oracle*, the two error terms separate on a *fixed* operator with no model fit, and the resulting sign rule can be tested *a priori* and *bidirectionally*; the pKa domain supplies exactly that test as a certified, in-distribution sign-flip (§5.7). This oracle-on-a-fixed-operator affordance is what the concept-leakage literature lacks[15,16], where the intermediate has no ground truth and leakage is inferred only indirectly. The solubility results—closure-dominance (§5.2) and the compensating surrogate (§5.4)—are the instrument's first application; we grade them *likely* and do not claim their transfer out of the matched regime. We therefore rest the paper on the instrument and its certified cross-domain demonstration, and report the solubility attribution at its measured strength.

**The accuracy case, and its boundary.** That the physics prior does not improve accuracy has no confirmed exception: the grounded model trails the matched black box on the full corpus, the external SOTA (FastSolv, SolProp) does no better on a difficulty-bounding by-solute split (App. E), and grounding the crystal factor on external labels did not move the ceiling (§G.3). The one place the physical *structure* helps is temperature extrapolation—but as a hypothesis-class property the trained models do not exploit (§G.1). These point to a *structural* ceiling: the crystal/activity split is non-identifiable from solubility alone (Lemma 1), and the activity closure is misspecified where we can measure it (§5.2), so within this family better inputs, more data, and auxiliary grounding do not recover an accuracy the factorization cannot express. What a misspecified closure costs is thus accuracy and, more consequentially, the interpretability the bottleneck appeared to buy.

**When a fixed-closure prior helps versus hurts.** The penalty sign rule (Prop. 1) predicts *when* to expect these negatives: a fixed-closure prior helps only when the estimation variance it saves exceeds its asymptotic bias $\Delta_\infty$ (both measurable). The "grounding helps iff the closure is well specified" reading is thus a statement about the *bias* channel ($\Delta_\infty$); even a correctly specified or true-valued input need not win on total error, because a fitted nuisance can be the more *efficient* estimator—the semiparametric phenomenon by which an estimated propensity beats the known-true one[45], sharpest where plug-in efficiency is delicate[46]—and Prop. 1 already carries this, its variance term overturning the sign the bias term alone would predict. The variance benefit is largest at small $n$ and vanishes as data accumulate or as the black box nears the aleatoric floor, whereas $\Delta_\infty$ is set by the closure; a prior is therefore predicted to hurt precisely in the mature-data, low-noise, well-covered regime. The synthetic sweep (§5.5) and the pKa closure (§5.7) exercise the fidelity and sufficiency behaviour directly. Whether grounding helps is shaped by *two* axes—closure fidelity and how the latent is trained (supervised vs. end-to-end); characterising that plane in full, and sweeping each axis directly on real systems, is deferred to future work.

**The novelty inversion (exploratory).** The sharpest—and most unexpected—confirmation of the penalty sign rule runs opposite to the usual coverage intuition. Resolving the asymptotic physics penalty by solute novelty, the closure is neutral on the most *novel* solutes (nearest-train Tanimoto $\leq 0.3$: $\Delta\mathrm{MAE} = -0.04 \pm 0.18$) and heaviest on the most *familiar*, best-covered ones (Tanimoto $> 0.8$: $+0.51 \pm 0.09$)—exactly as Proposition 1 predicts, since the variance a prior saves vanishes precisely where the black box is already accurate, leaving a structural closure bias as the dominant residual. The penalty likewise concentrates in strongly hydrogen-bonding solvents (water $+0.52 \pm 0.35$) and reverses sign in the acceptor-dominated amides ($-0.22 \pm 0.11$) the closure was built for (§G.4). Both patterns are exploratory—no multiplicity control across strata families—but they land where the penalty sign rule predicts, and where much of modern machine learning operates. We claim no universal boundary; we *propose*, but do not here cleanly validate, a diagnostic for the sign of the asymptotic physics penalty: measure $B_{\mathrm{closure}}$ through the decomposition, read $\Delta_\infty$ off the large-$n$ asymptote, and the sign should follow. In this sense the failure is not idiosyncratic to solubility; it recurs whenever a misspecified closure is coupled to an expressive encoder on abundant, low-noise data.

**Limitations.** (i) The phenomenon and its causal attribution live in *different regimes*: the paradox is a finite-composition solubility effect, but the closure–insufficiency measurement is made on the infinite-dilution/298 K/low-$\gamma$ VT-2005-matchable regime, so the transfer of the attribution back to the paradox's own regime is argued from in-regime signs, not measured there (§5.8); moreover, because few drug-like solutes have a VT-2005 reference profile, the solubility substitution falls almost entirely on the solvent channel, so the clean both-channel evidence is on the activity axis rather than the solubility axis. The attribution that the misspecification term is the larger rests on a one-sided bound with clustered-bootstrap support $P_{\mathrm{boot}} \approx 0.78$ on $n$=60 matched pairs, which is below conventional statistical significance. (ii) $B_{\mathrm{insuff}}$ is upper-bounded, not point-identified (no replicates at fixed $z^\star$); the smoothness-free Jensen bound assumes $m$ and $g$ share the $\ln \gamma_2^\infty$ reference state, which we verify (§5.2), so the mean offset is genuine decoder bias, not a units artifact. (iii) The constant-offset margin is convention-specific and inverts under the deployed residual-only default; there the separation rests on the convention-independent bound and the more-physics-worse effect. (iv) Absolute scaffold accuracy is encoder-bound but well *above* the inter-lab noise floor (0.15–0.31 in $\ln x_2$; §G.1), so label noise does not explain the ceiling. (v) The compensating-surrogate result is three-seed mean±sd ($n$=44; §5.4); its load-bearing content is the *sign* and low-rank, transferable *structure*, with the magnitudes ($51 \pm 2\%/72 \pm 1\%/4.1 \pm 0.5\times$) reported but varying across pipeline versions. (vi) The closure-fidelity contrast is measured on the activity axis; we have *not* tested whether the *solubility* paradox itself survives a modern 2010/dsp closure. That test needs a typed (donor/acceptor) $\sigma$-profile head and a UD $\cap$ BigSolDB matched set, and is the natural next experiment.

## 7 Conclusion

Physics-informed models are attractive because they promise both accuracy and interpretability, on the assumption that grounding their internal quantities in reference physics can only help. We have shown that this assumption fails when the physical model that consumes those quantities is misspecified. The learned intermediate then does not converge to the physical quantity it represents; it becomes a surrogate that compensates for the model's error, so supplying more faithful inputs removes the compensation and degrades the prediction. The failure is diagnosable: an external reference for the intermediate separates model misspecification from input insufficiency and attributes the accuracy ceiling to the fixed model.

The practical consequence is that grounding a learned latent on reference values, and reading that latent as the physical quantity, is justified only to the extent that the downstream physical model is faithful. When it is not, neither more accurate inputs nor a downstream correction lifts the ceiling; the remedy is a better model. The instrument we introduce makes this condition measurable for any predictor that composes a fixed physical model over a learned intermediate, and we expect the same compensating-surrogate behaviour wherever such a composition is trained end to end through an imperfect model.

## Author contributions

N. L. Polomoshnov: `[CRediT roles, e.g. Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing -- original draft]`. `[Co-author name]`: `[roles, e.g. Supervision, Funding acquisition, Project administration, Writing -- review & editing]`.

## Conflicts of interest

There are no conflicts to declare.

## Data availability

All code and the intermediate result artifacts needed to regenerate every figure, table, and reported number are available at `[repository URL]` (commit `[hash]`), released under `[license]`; the large model checkpoints and full per-arm prediction files are deposited at `[Zenodo DOI]`. Every data source is public: the solubility corpus is BigSolDB 2.0[29], the reference $\sigma$-profiles are the VT-2005 database, the experimental infinite-dilution activity coefficients are compiled from ThermoML, and the second-domain pKa data are the OPERA compilation[44]. The repository documents the exact command that reproduces each figure, table, and number from these sources, including the Bemis–Murcko scaffold split, the controlled grounded-vs-free comparison, the closure–insufficiency decomposition over the $n$=60 VT-2005-matchable pairs, the pKa second-domain measurement, and the synthetic fidelity sweep. The closure-fidelity contrast (§5.2) reports the *fair* full-2002 error (1.757) from our in-house NIST-validated COSMO-SAC-2002 layer, each closure evaluated on its prescribed native averaging; `run_fidelity_lever_fair2002.py` regenerates both the corner ($1.757 \to 0.765$) and the representative ($n$=477, $1.911 \to 1.855$) contrasts and asserts they reproduce these values (with `cCOSMO` cross-checks), while the earlier deposited `cCOSMO` flip artifact additionally carries the crippled single-channel 2002 diagnostic (1.263). Two artifacts carry disclosed caveats (§I): the seed-44 oracle per-row file is truncated (the run-time three-seed aggregates are unaffected), and the three-seed surrogate run's per-run split provenance was not logged (its matched set is the committed $n$=44).

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# Supplementary Information

## A Notation

## B Supplementary methods

**COSMO-SAC constants.** The differentiable COSMO-SAC layer uses the standard VT-2005

Table 2: Notation and coined terms used throughout.

| Symbol / term | Meaning |
|---|---|
| $g$ (closure) | fixed, non-fitted physical map (COSMO-SAC; a Hammett LFER for pKa) |
| $h$ (encoder) | learned graph network, $x \mapsto z$ |
| $z^\star$, $\hat{z}$ | reference (oracle) vs. learned physical intermediate (the $\sigma$-profile) |
| $m$ | activity target (experimental $\ln \gamma_2^\infty$) |
| $B_{\text{closure}}$ | closure misspecification, $\mathbb{E}[(\mathbb{E}[m \mid z^\star] - g(z^\star))^2]$ |
| $B_{\text{insuff}}$ | input insufficiency, $\mathbb{E}[\mathrm{Var}(m \mid z^\star)]$ |
| $\Gamma$ (grounding gap) | $R_{\text{orc}} - R_{\text{e2e}}$: oracle risk minus best end-to-end risk |
| $\Delta_\infty$ (physics penalty) | asymptotic risk of the physics arm minus the matched control |
| full / res | COSMO-SAC with / without the Staverman–Guggenheim combinatorial term |
| $R_{\text{orc}}, R_{\text{e2e}}, R_{\text{free}}$ | oracle / best end-to-end / free black-box population risks (they define $\Gamma$) |
| compensating surrogate | a learned $\hat{z}$ that *cancels* the closure's error instead of matching $z^\star$; here low-rank and out-of-sample transferable |
| $P_{\text{boot}}$ | clustered-bootstrap frequency of $B_{\text{closure}} > B_{\text{insuff}}$ on the matched set (a resampling frequency, not a posterior probability) |

/ Lin–Sandler constants: effective segment area $a_{\text{eff}} = 7.5\,\text{Å}^2$, electrostatic-misfit constant $\alpha' = 16466.72\,\text{kcal}\,\text{Å}^4\,\text{mol}^{-1}\,\text{e}^{-2}$, hydrogen-bond coefficient $c_{\text{hb}} = 85580$ with cutoff $\sigma_{\text{hb}} = 0.0084\,\text{e}\,\text{Å}^{-2}$, coordination number $z = 10$, and Staverman–Guggenheim normalisers $r_0 = 66.69\,\text{Å}^3$, $q_0 = 79.53\,\text{Å}^2$; the $\sigma$-profile grid is 51 bins on $[-0.025, 0.025]\,\text{e}\,\text{Å}^{-2}$. The segment-activity fixed point is damped ($\lambda = 0.7$) and run for 16 iterations at train time and 30 at evaluation.

**A usage pitfall in `cCOSMO` (`COSMO1` on typed profiles).** Handing the 2002 kernel a typed three-profile base produces a silently-wrong result that the library returns without warning. The reference implementation[38] pairs its 2002 model (`COSMO1`) with the single-profile Virginia Tech database; the demonstrated input is one untyped, Mullins-averaged profile. When `COSMO1` is instead handed a typed three-profile (Hsieh, `sigma3`) base — tempting when a 2002 and a 2010 arm are to share one profile source — it silently reads the nonpolar channel only, discards the donor and acceptor channels, keeps the total cavity area as prefactor, and returns a number without error. The prescribed way to recover a single 2002 profile from the typed set is instead their *sum*, $p = p_{\text{nhb}} + p_{\text{OH}} + p_{\text{OT}}$ (Eq. 9 of Ref.[38]); the nonpolar-only read is neither that sum nor a documented mode. This single-channel arm scores MSE 1.263 on the matched corner, whereas the fair full 2002 (the same molecule fed either to our differentiable layer or to `cCOSMO` with the full profile placed in the single nonpolar channel it actually reads) scores 1.757/1.758. This is an issue for the library maintainers (a silently-wrong return where an error would be safer), not a defect of any published benchmark.

**Crystal head.** $T_m = 100 + 600\,\sigma(\cdot) \in [100, 700]\,\text{K}$, $\Delta H_{\text{fus}} = S_H\,\text{softplus}(\cdot)$; the group-contribution $T_m$ prior is affine- calibrated on the train split before entering the head.

**Architecture.** The headline COSMO-SAC model uses a message-passing encoder (a shared backbone with a 2-layer residual role-specific branch) of hidden width 64 over 3 message-passing layers, on 35-dimensional node and 8-dimensional edge features; a set2set readout (3 processing steps) pools each graph, solute and solvent interact through a 3-layer cross-attention block, and the 512-dimensional pair representation $[\,g_s, g_v, g_s \odot g_v, |g_s - g_v|\,]$ (dropout 0.012) feeds the crystal head, the 51-bin $\sigma$-profile head, and the parameter-free COSMO-SAC closure. The full model has 5.40M trainable parameters—capacity that the in-sample linear probe ($R^2 \approx 0.76$, §G.1) indicates is not the binding constraint. The DirectGNN control shares this encoder, interaction, and readout verbatim.

**Encoder backbones.** The shared encoder has three interchangeable backbones: the headline message-passing network (MPNN); a hybrid local-message/global-attention graph transformer (GPS[47]); and a physics-channelled variant, Thermodynamically-Informed Message Passing (TIMP), which splits messages into dispersive, polar, and hydrogen-bonding channels supervised toward the Hansen parameters to prevent channel collapse. TIMP does not beat the plain MPNN across our runs, consistent with the encoder being transfer-limited rather than expressivity-limited (linear probe train $R^2 \approx 0.76$, test $R^2 \approx 0.45$).

**Optimisation.** Adam (weight decay $5 \times 10^{-4}$, gradient clipping at 2.0, batch 64) under the three-phase curriculum of §4: 30 epochs of auxiliary pretraining (P1), 70 of full SLE training (P2), and 10 of low-rate stabilisation (P3), at learning rates $2.6 \times 10^{-4}$, $8.5 \times 10^{-5}$, and $8.5 \times 10^{-6}$ respectively.

**Loss.** The objective is a weighted sum. Of its 32 registered terms, 17 carry weight 0 by default and never enter training—disabled experimental scaffolding (TIMP channel supervision; van't Hoff / temperature terms, $\times 7$; Hansen-contrastive terms, $\times 5$; descriptor/group priors;

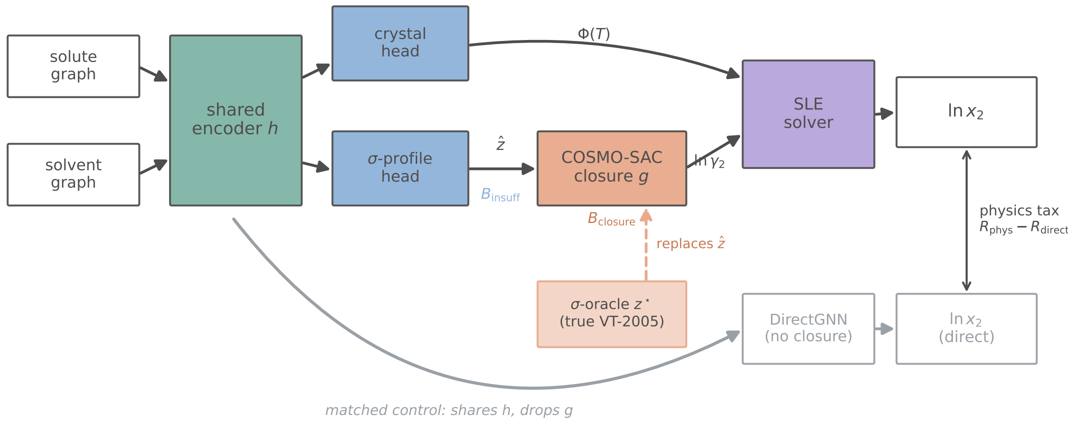


Figure 6: The model and the oracle substitution. Solute and solvent graphs pass through a shared encoder to a crystal head (the ideal term $\Phi(T)$) and a $\sigma$-profile head; the predicted profile $\hat{z}$ feeds the fixed COSMO-SAC closure $g$, which returns $\ln\gamma_2$, and an SLE solver combines the two into $\ln x_2$. Input insufficiency $B_{\text{insuff}}$ enters at the intermediate $\hat{z}$ and closure misspecification $B_{\text{closure}}$ at $g$. The $\sigma$-oracle replaces $\hat{z}$ with the reference VT-2005 profile $z^\star$ (salmon). DirectGNN (grey) is the matched control—it shares the encoder but drops the closure.

an auxiliary direct-solubility term), released with the code for completeness. The 15 *active* terms are below; only the top two groups are load-bearing for the claims, and a loss-simplification ablation (§B) tests which.

| role | term | default wt. |
|---|---|---|
| task | solubility (bin-weighted Huber on $\ln x_2$) | 1.0 |
| crystal grounding | melting point $T_m$ | 0.3 |
| | fusion enthalpy $\Delta H_{\text{fus}}$ | 0.3 |
| $\sigma$/act. ground. | infinite-dilution $\ln\gamma_2^\infty$ | 0.5 |
| | finite-composition $\ln\gamma_2$ | 0.5 |
| | Hansen parameters | 0.2 |
| | $\sigma$-profile EMD (grounding stream on) | † |
| regulariser | solubility monotonicity in $T$ | 0.1 |
| | physics preference | 0.1 |
| | adaptive-correction residual | 0.01 |
| | NRTL $\tau$ | 0.01 |
| | solubility variance | cfg |
| control coupling | DirectGNN NLL | 0.2 |
| | DirectGNN regulariser | 0.05 |
| architecture | mixture-of-experts balance | 0.02 |

Table 3: The 15 active loss terms (non-zero default weight). †The $\sigma$-profile earth-mover term is active only when the grounding stream is on. A further 17 registered terms are disabled (weight 0). Per-phase weight overrides are pinned in each run's resolved configuration.

**Loss-simplification ablation.** For a diagnostic result the elaborate objective is a potential confound: it could be read as indicating that the paradox is an artifact of loss tuning. We test the opposite—that the paradox and the compensating-surrogate drift survive a *minimal* objective. Pre-registered before the run: train the end-to-end grounded arm with only the task ($\ln x_2$ Huber), the crystal grounding ($T_m, \Delta H_{\text{fus}}$), and the $\sigma$-profile warmup, *dropping* the activity-grounding auxiliaries ($\ln\gamma_2^\infty$, $\ln\gamma_2$, Hansen) and every regulariser, then read the sign of the paradox (oracle vs. learned) and the surrogate drift. Prediction: both persist, because the mechanism is end-to-end training through a misspecified closure, not the auxiliary terms. Confirmed at the level the ablation measures. The ablation re-measures the solubility *paradox* (reference minus learned $\ln x_2$ MAE), the observable effect, not the profile drift directly: on the full test set it is $+1.13$ (95% CI $[0.88, 1.38]$) under the minimal objective, *larger* than the $+0.30$ ($[0.20, 0.39]$) under the full one. This is the *overall*, solvent-dominated paradox, and it strengthens once the activity-grounding auxiliaries—the only pressure toward physics—are removed; on the both-reference subset ($n$=297 rows over the $\approx$7 VT-covered solutes) the sign in fact reverses ($-0.23$ minimal vs $+0.22$ full), so the ablation is evidence that the overall observable effect is loss-robust, not subset-level confirmation of the mechanism. It is *consistent with* the compensating surrogate being intrinsic to end-to-end training through a misspecified closure rather than a regularisation artifact; it is an implication of the amplified paradox, not a direct re-measurement of $\|\hat{\sigma} - \sigma\|$ under the minimal run (the minimal-loss checkpoint was not retained, and the larger reference-arm error is consistent with more profile drift but does not isolate it from closure-path recalibration). The minimal model trains normally (validation $\ln x_2$ MAE 1.87, within 0.1 of the full-loss base), ruling out a degenerate-training reading (seed 42; `scripts/analysis/run_loss_ablation_analysis.py`).

# C Proofs

*Proof of Lemma 1 (non-identifiability).* Write $u = 1/T$ and $h = \Delta H_{\text{fus}}/R$, $u_m = 1/T_m$. At infinite dilution and with $\Delta C_p = 0$ the observable is

$$\mu(u) = -\Phi(u) - \ln \gamma_2^\infty(u) = -h(u-u_m) - (c + d\,u) = -(h+d)\,u + (hu_m - c),$$

using $\ln \gamma_2^\infty(u) = c + d\,u$ (affine in $u$ by hypothesis). Thus $\mu$ depends on the parameters $\theta = (\Delta H_{\text{fus}}, T_m, c, d)$ only through the pair $(S, K) = (-(h+d),\, hu_m - c) \in \mathbb{R}^2$. The activity block spans this plane, $\partial(S,K)/\partial c = (0, -1)$ and $\partial(S,K)/\partial d = (-1, 0)$; hence for *any* crystal value $(h, u_m)$ and any target $(S, K)$ there is an activity pair $d = -S - h$, $c = hu_m - K$ reproducing it. The likelihood level set therefore contains the full two-dimensional family $\{(h, u_m)\}$, so $(\Delta H_{\text{fus}}, T_m)$ are unidentified. Equivalently, the crystal score $\partial(S,K)/\partial h = (-1, u_m)$ lies in the span of the activity columns, so profiling out $(c, d)$ annihilates it and $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = 0$. At finite dilution $\ln \gamma_2(u) = \ln \gamma_2^\infty(u)\,(1-x_2)^2$ adds $\ln \gamma_2^\infty(u)\,(2x_2 - x_2^2) = O(x_2)$, which is not affine in $u$ because $x_2 = x_2(u)$; the crystal score then has a component outside the activity span of order $x_2$, so $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = O(x_2^2) > 0$. ☐

**Lemma 3** (Class-dependent efficient information)**.** *Let the activity nuisance lie in a class $\mathcal{H}$ with tangent projection $\Pi_{\mathcal{H}}$, and let $v = \partial\mu/\partial\Delta H_{\text{fus}}$ be the crystal score. The efficient information for $\Delta H_{\text{fus}}$ is $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = \sigma^{-2}\,\|v - \Pi_{\mathcal{H}} v\|^2$. If $\mathcal{H}$ contains the crystal score's span $\{1, 1/T\}$ (a free intercept and slope) then $\Pi_{\mathcal{H}} v = v$ and $\mathcal{I}_{\text{eff}} = 0$, recovering Lemma 1; if $\mathcal{H}$ is restricted (the slope is known or constrained) then $\mathcal{I}_{\text{eff}} > 0$ and the Cramér–Rao lower bound (CRLB) on $\Delta H_{\text{fus}}$ is finite. (The melting point $T_m$ is likewise an unknown crystal parameter, but is profiled jointly: its score $\partial\mu/\partial T_m$ lies in* $\operatorname{span}\{1\} \subset \operatorname{span}\{1, 1/T\}$*, already inside the nuisance tangent whenever the slope is free, so the dichotomy is unaffected by treating $T_m$ as free.)*

*Proof of Lemma 3 (class-dependent information).* This is the semiparametric efficient-score construction[48] (here finite-dimensional and Gaussian, so equivalently the partitioned-information / Frisch–Waugh–Lovell identity). With Gaussian noise of variance $\sigma^2$, parameter-of-interest score $v = \partial\mu/\partial\Delta H_{\text{fus}}$, and nuisance tangent space $\mathcal{T}_{\mathcal{H}}$ (the $L^2$-closure of the span of nuisance scores), the efficient score is the residual $\tilde{v} = v - \Pi_{\mathcal{H}} v$ and the efficient information is $\mathcal{I}_{\text{eff}} = \sigma^{-2}\|v - \Pi_{\mathcal{H}} v\|^2$. Here $v = \partial\mu/\partial\Delta H_{\text{fus}} = -(u - u_m)/R \in \operatorname{span}\{1, u\}$. If $\mathcal{T}_{\mathcal{H}} \supseteq \operatorname{span}\{1, u\}$ then $\Pi_{\mathcal{H}} v = v$ and $\mathcal{I}_{\text{eff}} = 0$, recovering Lemma 1; if $\mathcal{T}_{\mathcal{H}}$ is a proper subspace not containing $v$ (e.g. a fixed slope, $\mathcal{T}_{\mathcal{H}} = \operatorname{span}\{1\}$) then $\Pi_{\mathcal{H}} v \neq v$ and $\mathcal{I}_{\text{eff}} > 0$, with a finite Cramér–Rao bound. ☐

*Proof of Lemma 2 (decomposition and bounds). (a) Identity.* Split the residual as $m - g(z^\star) = [m - \mathbb{E}(m \mid z^\star)] + [\mathbb{E}(m \mid z^\star) - g(z^\star)]$, square, and take expectations. The squared terms give $\mathbb{E}[\operatorname{Var}(m \mid z^\star)] = B_{\text{insuff}}$ and $\mathbb{E}[(\mathbb{E}(m \mid z^\star) - g(z^\star))^2] = B_{\text{closure}}$. The cross term vanishes: since $\mathbb{E}(m \mid z^\star) - g(z^\star)$ is $\sigma(z^\star)$-measurable, the tower rule gives $\mathbb{E}\big[(m - \mathbb{E}(m \mid z^\star))(\mathbb{E}(m \mid z^\star) - g(z^\star))\big] = \mathbb{E}\big[(\mathbb{E}(m \mid z^\star) - g(z^\star))\,\mathbb{E}(m - \mathbb{E}(m \mid z^\star) \mid z^\star)\big] = 0$. *(b) Jensen lower bound.* With $\Delta(z^\star) = \mathbb{E}(m \mid z^\star) - g(z^\star)$, $B_{\text{closure}} = \mathbb{E}[\Delta^2] \geq (\mathbb{E}[\Delta])^2 = (\mathbb{E}[m] - \mathbb{E}[g])^2$ by Jensen, using $\mathbb{E}[\mathbb{E}(m \mid z^\star)] = \mathbb{E}[m]$; it requires only that $m$ and $g$ share a reference state, so that $\mathbb{E}[m] - \mathbb{E}[g]$ is a genuine mean discrepancy. *(c) Law-of-total-variance upper bound.* For any binning $B = \operatorname{bin}(g(z^\star))$, $\sigma(B) \subseteq \sigma(z^\star)$, so conditioning on the coarser $\sigma$-algebra cannot reduce expected conditional variance: $\mathbb{E}[\operatorname{Var}(m \mid B)] \geq \mathbb{E}[\operatorname{Var}(m \mid z^\star)] = B_{\text{insuff}}$. *(d) Convention-independence.* $B_{\text{insuff}} = \mathbb{E}[\operatorname{Var}(m \mid z^\star)]$ is a functional of the law of $(m, z^\star)$ alone and does not involve $g$; hence it is identical across closure conventions, and any $\operatorname{MSE}(g_1) - \operatorname{MSE}(g_2)$ is entirely $B_{\text{closure}}(g_1) - B_{\text{closure}}(g_2)$. ☐

**Theorem 1** (Grounding-gap sandwich)**.** *Assume all risks are finite ($m, g(z^\star) \in L^2$; $g$ and each $h \in \mathcal{E}$ measurable), (A1) the true encoder is representable, $\phi^\star \in \mathcal{E}$ with $z^\star = \phi^\star(x)$, and (A2) the intermediate is a lossless bottleneck, $\mathbb{E}[m \mid x] = \mathbb{E}[m \mid z^\star]$. Then*

$$0 \;\leq\; \Gamma \;\leq\; B_{\text{closure}}, \qquad \Gamma = B_{\text{closure}} - \big(R_{\text{e2e}} - R_{\text{free}}\big), \tag{6}$$

*with $\Gamma = B_{\text{closure}}$ iff $R_{\text{e2e}} = R_{\text{free}}$ (the composed class realizes the Bayes predictor by distorting the intermediate). In particular a well-specified closure ($B_{\text{closure}} = 0$) gives $\Gamma = 0$; hence* under A1–A2, *$\Gamma > 0$ certifies $B_{\text{closure}} > 0$.*

*Proof of Theorem 1 (grounding-gap sandwich).* Throughout assume $m, g(z^\star) \in L^2$ and $g$, every $h \in \mathcal{E}$ measurable. *(a) $\Gamma \geq 0$.* By A1, $\phi^\star \in \mathcal{E}$ is a feasible encoder with $g(\phi^\star(x)) = g(z^\star)$, so $R_{\text{e2e}} = \inf_h \mathbb{E}[(m - g(h(x)))^2] \leq \mathbb{E}[(m - g(z^\star))^2] = R_{\text{orc}}$; the infimum need not be attained. Hence $\Gamma = R_{\text{orc}} - R_{\text{e2e}} \geq 0$. *(b) $R_{\text{e2e}} \geq R_{\text{free}}$.* Every $g(h(x))$ is $\sigma(x)$-measurable, and $R_{\text{free}} = \mathbb{E}[(m - \mathbb{E}(m \mid x))^2]$ is the infimum of $\mathbb{E}[(m - p)^2]$ over *all* $\sigma(x)$-measurable $p$; an infimum over the subclass $\{g(h(\cdot)) : h \in \mathcal{E}\}$ is no smaller. *(c) $R_{\text{free}} = B_{\text{insuff}}$ under A2.* From $\operatorname{Var}(m) = \mathbb{E}[\operatorname{Var}(m \mid x)] + \operatorname{Var}(\mathbb{E}(m \mid x)) = \mathbb{E}[\operatorname{Var}(m \mid z^\star)] + \operatorname{Var}(\mathbb{E}(m \mid z^\star))$ and A2 ($\mathbb{E}(m \mid x) = \mathbb{E}(m \mid z^\star)$, so the two $\operatorname{Var}(\mathbb{E}(\cdot))$ terms agree), the two expected conditional variances agree: $R_{\text{free}} = \mathbb{E}[\operatorname{Var}(m \mid x)] = B_{\text{insuff}}$. *(d) Sandwich and identity.* By Lemma 2, $R_{\text{orc}} = B_{\text{insuff}} + B_{\text{closure}}$. With (c), $\Gamma = R_{\text{orc}} - R_{\text{e2e}} = B_{\text{closure}} - (R_{\text{e2e}} - R_{\text{free}})$, and (b) gives $R_{\text{e2e}} - R_{\text{free}} \geq 0$, so $0 \leq \Gamma \leq B_{\text{closure}}$; equality $\Gamma = B_{\text{closure}}$ holds iff $R_{\text{e2e}} = R_{\text{free}}$. Finally $B_{\text{closure}} = 0$ forces $0 \leq \Gamma \leq 0$, i.e. $\Gamma = 0$; contrapositively, under A1–A2, $\Gamma > 0 \Rightarrow B_{\text{closure}} > 0$. *(e) A2 is necessary for the certificate.* If A2 fails, take $x \in \{a, b\}$ equiprobable, $z^\star \equiv 0$ (so A1 holds), $m(a) = +1, m(b) = -1$, and $g$

with $g(0) = 0 = \mathbb{E}(m \mid z^\star)$ so $B_{\text{closure}} = 0$; an encoder $h^\dagger(a) = 1, h^\dagger(b) = -1$ with $g(\pm 1) = \pm 1$ gives $R_{\text{e2e}} = 0$ while $R_{\text{orc}} = 1$, hence $\Gamma = 1 > 0 = B_{\text{closure}}$. So $\Gamma > 0$ alone does not certify $B_{\text{closure}} > 0$; the decomposition (Lemma 2) does. ☐

The penalty-sign result (Proposition 1, stated in the main text at §3.3) is proved here.

*Proof of Proposition 1 (sign of the physics penalty).* Write $D(n) := V_{\text{direct}}(n) - V_{\text{phys}}(n)$ and $\Delta_\infty := R_{\text{e2e}} - R_{\text{direct},\infty}$, so that $\mathcal{T}(n) = \Delta_\infty - D(n)$. *(i)* $\mathcal{T}(n) < 0$ exactly when $D(n) > \Delta_\infty$, immediately. *(ii)* If $\Delta_\infty > 0$ and $D(n) \to 0$, choose $n^\star$ with $|D(n)| < \Delta_\infty$ for all $n > n^\star$ (possible since $D(n) \to 0$); then $\mathcal{T}(n) = \Delta_\infty - D(n) > 0$. Monotonicity of $D$ is *not* required for existence of $n^\star$; it is needed only for the stronger claim "$\mathcal{T}(n) > 0$ at every $n$," which we treat as an empirical observation. *(iii)* Since $V_{\text{phys}}(n) \geq 0$, $D(n) \leq V_{\text{direct}}(n)$: the variance a prior can save is bounded by the control's own estimation variance. As the black box approaches the aleatoric floor $V_{\text{direct}}(n) \to 0$, so $\limsup_n D(n) \leq 0$; with both estimators consistent, so $V_{\text{phys}}, V_{\text{direct}} \to 0$, one has $D(n) \to 0$ and $\mathcal{T}(n) \to \Delta_\infty \geq 0$. When the control is asymptotically Bayes, so $R_{\text{direct},\infty} = R_{\text{free}}$, and A2 holds, then $\Delta_\infty = R_{\text{e2e}} - R_{\text{free}} = B_{\text{closure}} - \Gamma$ by (6). ☐

# D Per-arm tables and decomposition robustness

Table 4 gives the exact per-arm metrics behind Fig. 2: three seeds (mean±sd) on the cross-arm intersection lock ($n$=5608, rows supervised and finite in every arm). Both closure families—NRTL and COSMO-SAC over a learned $\sigma$-profile—trail the DirectGNN control (the physics penalty); the $\sigma$-oracle (reference VT-2005 profile) is the worst arm, and restoring the Staverman–Guggenheim combinatorial term makes it slightly worse still—the signature of a misspecified map (§5.2).

**Robustness of the closure/insufficiency separation.** We stress the deployed-convention verdict $B_{\text{closure}} > B_{\text{insuff}}$ (§5.2)—equivalently $B_{\text{insuff}} < \text{MSE}/2 = 0.74$ at $\text{MSE}_{\text{res}} = 1.47$—four ways on the $n$=60 set.

*(i) Estimator sensitivity.* The four $B_{\text{insuff}}$ upper bounds of Table 1—0.40 ($k$-NN), 0.62 (binning/LOTV), 0.56 (random forest), 0.62 (ridge out-of-fold)—all clear the bar. The $k$-NN value is the smallest and least conservative input to $B_{\text{closure}} \geq \text{MSE} - B^{\text{up}}_{\text{insuff}}$, so we headline the leakage-immune LOTV bound instead. The LOTV bound bins $g(z^\star)$ and is therefore finite-sample downward-biased when points per bin are few, which would *inflate* $B_{\text{closure}}$ favourably; but the two non-binning bounds (RF 0.56, ridge 0.62) are immune to that per-bin sparsity and reproduce 0.62, so the headline does not depend on the binning estimator. The sole non-certifying configuration is a deliberately aggressive ridge, which lifts $B_{\text{insuff}}$ to 0.88 (then $B_{\text{closure}} \geq 0.59 < 0.88$): a sensitivity edge, not a refutation.

*(ii) Leave-one-solvent-out.* Dropping each of the 17 solvents in turn, the verdict holds in all 17 folds under the LOTV bound. Under the random-forest estimator it holds in only 2/17, because an RF over-fits conditional variance in 102 dimensions at $n$=60 (its out-of-fold MSE approaches $\text{Var}(m)$), so its leave-one-out estimate is unstable—an estimator artifact, not a sign flip.

*(iii) Label noise.* Injecting zero-mean Gaussian noise $\sigma_\eta \in \{0.2, \ldots, 0.8\}$ into $m$ leaves the $B_{\text{closure}}$ lower bound stable ($0.84 \to 0.70$), as expected since $B_{\text{closure}} = \text{MSE} - \mathbb{E}[\text{Var}(m \mid z^\star)]$ is invariant to zero-mean noise.

*(iv) Clustered-bootstrap interval.* A solvent-clustered bootstrap (3000 resamples of the 17 solvents, LOTV bound) makes the confidence explicit: the $B_{\text{closure}}$ lower bound has median 0.93 ($\text{CI}_{90}$ [0.40, 1.62]), and the margin $B_{\text{closure}} - B_{\text{insuff}}$ has median 0.36 ($\text{CI}_{90}$ [−0.22, +0.97]) with $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.89$ (single-axis; the conservative two-way solute×solvent bootstrap, which also charges for repeated solutes, gives $\approx 0.78$, the value the main text headlines). With $G = 17$ solvent and 41 solute clusters this small-$G$ regime has optimistic interval coverage, and the finite-cluster correction points the way we already conclude: less confidence, non-certification. We therefore treat the $P$ values as a coarse band, not two-decimal quantities. The separation is *likely* but not certified at the 90% level, consistent with the $\approx 0.85$ figure, which is why the main text treats it as supporting rather than decisive.

**In-regime ($\gamma$-stratified) separation.** The $n$=60 measurement is matched at low $\gamma$, while the paradox spans all $\gamma$. Stratifying the $n$=60 set by $\ln \gamma_2^\infty$ tests whether closure-dominance is a regime artifact. It is not: the verdict $B_{\text{closure}} > B_{\text{insuff}}$ (leakage-immune LOTV binning bound) holds in every stratum and *strengthens* toward higher $\gamma$ (Table 5)—in the highest stratum ($\overline{\ln \gamma_2^\infty} = 2.94$, approaching the non-ideal regime) $B_{\text{closure}} \geq 2.69$ exceeds $B_{\text{insuff}} \leq 0.67$ several-fold. We caution that the high-$\gamma$ strata are small ($n$ = 17–30 in 102 dimensions): the law-of-total-variance bound can be pushed toward zero by too-few-points-per-bin, which mechanically widens the margin exactly where data is thinnest, so it is the several-fold separations rather than the precise values that carry the qualitative reading. One boundary deserves an explicit note: the $m > \overline{m}$ stratum separates through its upper (high-$\gamma$) half, and the *isolated* moderate band ($\overline{m} < m \leq 1.5$) does *not* separate. This is not a counter-example. The total error there is only ≈0.06, so $B_{\text{closure}} \leq \text{MSE} \approx 0.06$ assumption-free (since $B_{\text{insuff}} \geq 0$), and there is simply nothing to attribute—near ideality the target is small and the

Table 4: Per-arm scaffold-split metrics (3 seeds, $n$=5608 cross-arm lock). Ordered by MAE.

| Arm | MAE | $R^2$ |
|---|---|---|
| DirectGNN (no closure) | $1.70 \pm 0.03$ | $+0.42 \pm 0.03$ |
| NRTL closure | $1.80 \pm 0.07$ | $+0.34 \pm 0.03$ |
| COSMO-SAC, learned $\sigma$ (grounded) | $1.85 \pm 0.05$ | $+0.33 \pm 0.05$ |
| + Staverman–Guggenheim combinatorial | $1.88 \pm 0.09$ | $+0.32 \pm 0.05$ |
| COSMO-SAC, ungrounded $\sigma$ | $2.04 \pm 0.04$ | $+0.19 \pm 0.01$ |
| COSMO-SAC, reference $\sigma$ ($\sigma$-oracle) | $2.25 \pm 0.02$ | $-0.03 \pm 0.04$ |

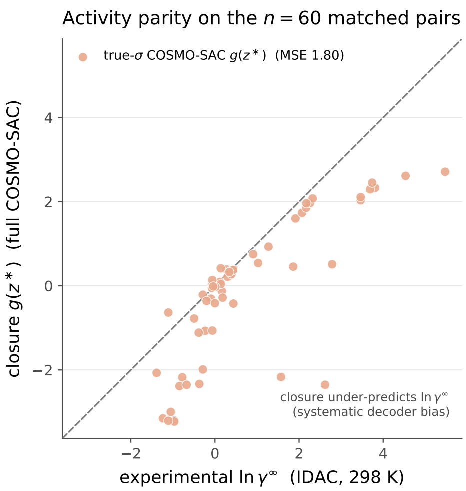


Figure 7: Activity-coefficient parity on the $n$=60 matched pairs. The reference-$\sigma$ COSMO-SAC closure $g(z^\star)$ (salmon) falls below the diagonal at larger $\ln\gamma_2^\infty$, under-predicting the activity coefficient (MSE 1.80, full convention).

split is uninformative. We read no closure-quality claim from it in either direction. This supports—but does not prove—the conjecture that the same closure misspecification drives the all-$\gamma$ paradox.

Table 5: Closure/insufficiency verdict across $\ln\gamma_2^\infty$ strata of the matched set (deployed residual-only convention; $B_{\text{insuff}}^{\text{up}}$ is the leakage-immune LOTV binning bound throughout).

| Stratum | $n$ | $\overline{\ln\gamma_2^\infty}$ | $B_{\text{insuff}} \le$ | $B_{\text{closure}} \ge$ |
|---|---|---|---|---|
| lower $\gamma$ ($m \le$ med.) | 30 | $-0.40$ | 0.08 | 0.95 |
| higher $\gamma$ ($m >$ med.) | 30 | $+1.90$ | 0.86 | 1.07 |
| high $\gamma$ ($m > 1.5$) | 17 | $+2.94$ | 0.67 | 2.69 |

The closure/insufficiency decomposition (full/res conventions; law-of-total-variance (LOTV), random-forest (RF), ridge, and kNN estimators; solvent-clustered bootstrap; the pyridine stratum) is tabulated in §5.2.

**The four hedges (full).** The adversarial audit could not invert the verdict but sharpened its scope:

1. *Convention.* The deployed default is residual-only. Under it the *constant-offset* (Jensen) lower bound inverts ($B_{\text{closure}} \ge 0.43 < B_{\text{insuff}} \le 0.62$; it separates only under the full convention, $B_{\text{closure}} \ge 0.72$), so that assumption-free bound does not separate under deployment (this is the convention-specific row deferred here from Table 1 / Fig. 3), and the full-convention margin (the sharpened MSE $- B_{\text{insuff}}$ bound, 1.27 vs 0.53) is a full-COSMO-SAC-only result. What separates under the deployed convention is the leakage-immune bound $B_{\text{closure}} \ge \text{MSE}_{\text{res}} - B_{\text{insuff}}^{\text{LOTV}} \approx 0.85 > 0.62$, on which we headline, together with the combinatorial Tier 2 fact.

2. *Confound.* The closure error is predominantly a *between-solvent* systematic bias (~64% of total error); within the largest single-solvent stratum (pyridine, $n$=20) the closure is nearly correct (stratum MSE 0.07). Because solvent $\sigma$ is part of $z^\star$, these offsets are still decoder error and give an independent solvent-conditional lower bound $B_{\text{closure}} \ge 0.89$ (full)/0.56 (res).

3. *Confidence.* Honest confidence on the full-convention constant-offset separation is the solvent-clustered bootstrap $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.85$ (0.83 vs the random-forest upper bound, 0.89 vs the law-of-total-variance bound); we report the solvent-clustered scheme rather than the less conservative i.i.d.-pair one because solvent dependence dominates. The leakage-immune margin (0.85 vs 0.62) is a point estimate. The decomposition is checkpoint/seed-independent (the closure has zero learnable parameters).

4. *Selection/scope.* The 60 matchable pairs are the low-to-moderate $\gamma$ regime (matched mean $\ln\gamma_2^\infty = 0.75$ vs 3.56 for VT-2005-unmatchable pairs, $t = -7.44$; the strongly non-ideal amide-solvent regime is untested because VT-2005 lacks those profiles). This narrows scope but is *conservative*, since the untested regime is a harder closure test. The target is $\ln\gamma_2^\infty$ at 298 K, so the numeric ceiling does not necessarily transfer unchanged to the finite-composition solubility regime.

# E External baselines and ablations

We retrained the direct SOTA (FastSolv[3]) and the physics-informed SOTA (SolProp[2]) on our corpus and evaluated them on a by-solute test split (`test_solute.csv`, $n \approx$ 10–12k). In $\ln x_2$ they reach MAE 1.94 (FastSolv, $R^2$ 0.23) and 2.34 (SolProp, calibrated, $R^2$ −0.18); in $\log_{10} S$, 0.87 and 1.04. These are on a *by-solute* split that is *not comparable* to our scaffold numbers (Table 4): they bound the task's difficulty, not a head-to-head ranking, so this does *not* establish that our scaffold-split DirectGNN (1.70) beats them, and a like-for-like scaffold evaluation of the external models is left to the deposited artifacts. What *is* comparable is the internal ordering: the physics-informed SolProp trails the direct FastSolv—the same ordering we find between our physics closures and DirectGNN. The regime-level reading is that on this corpus no black box, ours or external, escapes the accuracy ceiling—consistent with the structural account of §6. Per-arm ablations (NRTL vs COSMO-SAC, combinatorial on/off) are in Table 4.

# F Supplementary mechanism results

**Non-identifiability, empirically (the corrupted twin).** Training a twin model on data whose crystal labels are corrupted (every $T_m$ shifted by +273 K) leaves the *solubility* fit essentially unchanged—$\ln x_2$ MAE 1.835 versus 1.812 on the corrected data—while the recovered crystal grounding differs by hundreds of kelvin ($T_m$ MAE 258 K versus 45 K). Two very different crystal/activity splits produce the same solubility, which is Lemma 1 made concrete.

**Crystal grounding on physical labels.** When the crystal head is supervised on measured melting points it predicts $T_m$ to 45 K MAE, about 4× better than a Joback group-contribution reference—so the flat downstream result of §G.3 is a dose/identifiability effect, not an inability to learn crystal properties.

**The $\Delta C_p = 0$ approximation.** The ideal term omits the constant-heat- capacity correction; a group-contribution audit finds a nonzero $\Delta C_p$ for 1441 of 1525 solutes (108,287 rows), so this is a bounded model-form approximation. It cannot produce either headline result. The closure/insufficiency decomposition is measured on the activity target ($\ln \gamma_2^\infty$ at 298 K), where $\Delta C_p$ does not enter at all; and the grounding paradox is an arm-to-arm difference (learned-$\sigma$ vs. oracle-$\sigma$) whose two arms share the *identical* crystal term, so any $\Delta C_p$ error cancels exactly in the comparison. It is thus orthogonal to the activity closure, not a source of the paradox.

**Fusion-supervision scarcity.** Direct $\Delta H_{\text{fus}}$ labels are extremely sparse: 1279 training rows across 31 solutes, and essentially none in validation or test. This is the quantitative backbone of the identifiability argument—there is almost no signal with which to pin the crystal term from data.

**The NRTL branch trains (a trainability check).** The asymptotic physics penalty is not an undertraining artifact of the activity branch. In the headline NRTL arm the head produces solvent-varying interaction parameters—$\tau_{12}$ over the $n$=8103 test pairs has mean 0.97 and std 2.5 (range $[-2.6, 22]$), far from collapsed—yet the arm still trails the closure-free DirectGNN (1.80 vs 1.70; Table 4). A trained, non-degenerate activity branch beaten by the control is evidence that the penalty is structural, not a dead branch. (The core closure-misspecification result—the $\sigma$-oracle and the decomposition—is in any case training-*independent*: it feeds the reference VT-2005 profiles through the fixed closure with no encoder optimization, so it cannot be a training pathology.)

**Structured temperature extrapolation.** Withholding *temperature* rather than structure (train at $T \leq$ 309.75 K, test at higher $T$; $n = 3343$), the affine-in-$1/T$ hypothesis classes extrapolate far better than an unstructured descriptor regressor (Table 6); the van't Hoff class also predicts the *direction* of the high-$T$ shift correctly 99% of the time versus 40% for the random forest. This is a property of the structured hypothesis class—the trained neural models are weaker than the van't Hoff baseline—not an accuracy claim for our model.

Table 6: Same-pair high-temperature extrapolation ($n =$ 3343, train $T \leq 309.75$ K). "dir." is the fraction of pairs whose high-$T$ shift direction is predicted correctly.

| Model | MAE | $R^2$ | dir. |
|---|---|---|---|
| per-pair van't Hoff ($1/T$) | 0.37 | 0.89 | 0.99 |
| per-pair linear in $T$ | 0.41 | 0.85 | 0.99 |
| per-pair last-low-$T$ hold | 1.20 | 0.69 | 0.01 |
| RF (Morgan $+T$) | 1.29 | 0.66 | 0.40 |
| per-pair mean | 1.46 | 0.56 | 0.01 |

# G Supplementary diagnostics and broad negatives

## G.1 Supporting diagnostics

The interpretive claims of the preceding sections are based on a set of controlled diagnostics, each run under a named protocol. We report them here with their exact numbers, including the results that are unfavorable to the physics-informed hypothesis.

**Aleatoric noise floor.** Repeated solubility measurements of the same solute in the same solvent disagree across laboratories, and that disagreement sets a floor below which no model can be expected to fit. Measuring inter-source spread on[29], the mean absolute disagreement is approximately 0.15 $\ln x_2$ units for pairs backed by two sources ($n = 3948$ groups), rising to 0.31 for groups backed by three or more sources ($n = 349$); the corresponding medians are smaller still (0.03 and 0.09). The scaffold-split MAE we report ($\approx 1.7$ to 1.9) sits well above this band, so label noise does not account for the accuracy ceiling. The floor is small; the ceiling is not a data-quality artifact.

**The encoder is transfer-limited, not expressivity-limited.** To separate what the frozen graph embedding can represent from what it transfers out of distribution, we fit a linear probe from the encoder representation onto RDKit descriptors and compare in-distribution and held-out fit. The median train $R^2$ is 0.758 against a test $R^2$ of 0.448 ($n = 700$ solutes in each partition), a gap of 0.31. Individual descriptors split sharply: FractionCSP3 transfers well (test $R^2 = 0.96$) while HeavyAtomCount does not (0.23). The embedding is expressive enough in sample; what degrades out of distribution is the generalization of that expressivity across scaffolds. We read this as suggestive, not decisive: the sharp per-descriptor spread (FractionCSP3 0.96 vs HeavyAtomCount 0.23) means the train–test gap partly reflects covariate shift in the descriptor *targets* across scaffolds rather than an encoder property alone, and the probe measures descriptor-representability, not task-relevant expressivity directly. It therefore *suggests*—rather than establishes—that a new graph encoder is unlikely to be the main lever for out-of-distribution accuracy, with pretraining and coverage the more likely ones.

**Temperature extrapolation.** A structured van't Hoff activity class (log-solubility affine in $1/T$) encodes the correct temperature dependence of $\ln \gamma_2$ by construction, and this shows up when the split withholds temperature rather than structure. Training on measurements at $T \leq 309.75$ K and testing at $T \geq 330$ K ($n_{\text{test}} = 3343$), the structured class attains a high-temperature MAE of 0.37 against 1.29 for a random forest over Morgan fingerprints plus temperature (in $\ln x_2$ units), consistent with the temperature-aware $\ln \gamma_2^{\infty}$ modelling of[5]. This is a property of the structured hypothesis class, not a performance claim we make for the trained model; the point is that the functional form carries temperature extrapolation that a descriptor-plus-temperature regressor does not.

Table 7: Solvent-ranking quality of the physics path on the scaffold test ($n = 826$ solute groups).

| Metric | Value |
|---|---|
| Spearman $\rho$ | 0.511 |
| Top-1 accuracy | 0.477 |
| Kendall $\tau$ | 0.436 |
| nDCG@3 | 0.732 |

Table 8: Fisher audit of the SLE design at $\Delta T = 40$ K (synthetic, closure-independent).

| Supervision regime | cond($I$) | CRLB sd($\Delta H_{\text{fus}}$) [J/mol] |
|---|---|---|
| Solubility only | $2.3 \times 10^5$ | 19,895 |
| + $T_m$ label | $5.4 \times 10^3$ | 3,151 |
| + $T_m + \Delta H_{\text{fus}}$ | $1.6 \times 10^3$ | 1,689 |

**Ranking and calibration.** Absolute error and decision quality diverge. Despite mediocre MAE, the physics path orders candidate solvents usefully (Table 7; nDCG@3 $\approx 0.73$ across $n = 826$ solute groups). Uncertainty, by contrast, is badly miscalibrated in the native model—a nominal-90% interval covers only PICP@90 = 0.13 of held-out points—but split-conformal wrapping repairs it to 0.92 empirical coverage at the same nominal level. The ranking signal and the post-hoc calibrated intervals are the practically usable outputs, even where the point predictions are not competitive.

## G.2 Identifiability and compensation

Lemma 1 asserts that the crystal/activity split is not recoverable from solubility data alone. Two diagnostics make this concrete. The first is a numerical Fisher audit of the SLE design, computed on synthetic data and independent of any particular activity closure. We evaluate the Fisher information matrix $I$ of the observation model at a reference operating point and read off the conditioning cond($I$) and the Cramér–Rao lower bound on the standard deviation of $\Delta H_{\text{fus}}$ under three supervision regimes. Table 8 reports the result at a temperature window of $\Delta T = 40$ K. With solubility measurements only, the design is severely ill-conditioned and the CRLB on $\Delta H_{\text{fus}}$ is roughly 20,000 J/mol, comparable to the quantity being estimated. Adding a direct $T_m$ label drops the conditioning by nearly two orders of magnitude and cuts the bound to about 3,200 J/mol; supplying $T_m$ and $\Delta H_{\text{fus}}$ together brings it to about 1,700 J/mol. The ill-conditioning is a property of the design rather than of noise level, and it is closed only by external single-component supervision, which is the mechanism the aux-stream pattern supplies.

Widening or narrowing the temperature window does not rescue the solubility-only case. A $\Delta T$ sweep moves the solubility-only conditioning from $1.7 \times 10^3$ at a wide window to $2.3 \times 10^5$ at 40 K and to $1.9 \times 10^{16}$ at 10 K, so multi-temperature data alone does not resolve the split; the two parameters remain confounded along the ideal-term direction. Restricting the activity slope rather than adding labels helps, but only partially: it cuts $\mathrm{sd}(\Delta H_{\mathrm{fus}})$ to 8,251 J/mol, still well above the label-supervised regimes. A second, model-side diagnostic requires more careful interpretation and is not part of the primary argument. On the supervised test set, with a group-contribution (Joback) crystal reference, the crystal and activity residuals are strongly anti-correlated ($\mathrm{corr}(\delta\Phi, \delta \ln \gamma_2) = -0.962$). This is an *accounting artifact*, not evidence of a learned split: the two residuals sum to the (negative) $\ln x_2$ fit residual, so their correlation tends to $-1$ as the fit improves, a permutation null that destroys any learned coupling still reproduces most of it, and the model's own factors barely co-vary. The large apparent crystal offset is likewise not a usable misgrounding metric — it is dominated by reference error, since the Joback reference is physically implausible on these drug-like polycyclics, and capping predicted $T_m$ at a physical bound collapses the offset from $\approx -7.1$ to $\approx -1.5$.[3] We therefore base the non-identifiability claim on the reference-free Fisher/CRLB audit above, and present the compensation figures in the SI (§F) as a cautionary diagnostic rather than a decomposition-quality measurement.

## G.3 Grounding the crystal factor: a dose-limited null

A natural test of whether the ceiling is inputs rather than closure is to ground a *different* factor—the crystal term—on abundant external data. We trained the identical model without and with an external single-component crystal-property pool of roughly 15,000 melting points, added through the auxiliary-stream mechanism after excluding any pool solute whose Bemis–Murcko scaffold appears in the validation or test split. The pool did not improve prediction of the crystal properties it supplies (melting-point MAE 45.0 → 46.6 K, effectively unchanged, if anything marginally worse), and downstream solubility did not improve either ($\ln x_2$ MAE 1.81 → 1.88; two nominally favorable but slight shifts, in $\ln x_2$ $R^2$ and the crystal offset $|\delta\Phi|$, do not amount to a gain). We characterize this as *dose-limited and inconclusive rather than a clean null.* In this run the crystal auxiliary stream was applied at a low dose (a small fraction of optimizer steps at low loss weight), so the model saw the external pool less than one full pass at the selected checkpoint, and a stream that did not even lower the MAE of the property it directly supervises cannot justify a strong conclusion. What it shows is that *this low-dose intervention* does not move the ceiling, not that external crystal labels cannot. We therefore do *not* read it through the activity-path $B_{\mathrm{closure}}/B_{\mathrm{insuff}}$ decomposition: the crystal "closure" is the near-exact ideal-solubility term $\Phi = (\Delta H_{\mathrm{fus}}/R)(1/T - 1/T_m)$, not a misspecified map, so the flat $T_m$ recovery is better read as the crystal/activity split's structural non-identifiability (Lemma 1) and the encoder's transfer limitation (§G.1) than as closure dominance. A properly dosed crystal-grounding test is left to future work.

## G.4 Chemistry-resolved structure of the asymptotic physics penalty

Across the full test set the two arms are separated by little more than label noise. On the 3-seed scaffold split the DirectGNN control reaches an $\ln x_2$ MAE of $1.70 \pm 0.03$, while the grounded physics arm (learned-$\sigma$ COSMO-SAC closure) reaches $1.85 \pm 0.05$, for a global $\Delta\mathrm{MAE} = \mathrm{MAE(physics)} - \mathrm{MAE(control)} = +0.14$ (positive meaning the closure adds cost).[4] At this granularity the physics bottleneck is a small, near-noise penalty on accuracy. The interesting structure appears only once the error is resolved along chemical axes, where the average penalty is an average over regimes that disagree in sign.

The solvent axis is the most pronounced. Binning test pairs by solvent class and recomputing $\Delta$MAE per seed (Table 9) shows the cost concentrated in solvents that hydrogen-bond strongly and directionally, precisely the regime in which the COSMO-SAC closure is least faithful and $|\ln \gamma_2|$ is largest. Water carries the heaviest penalty, and the effect decreases monotonically as the donor strength of the class softens. One class inverts the sign: amide solvents, which are aprotic dipolar hydrogen-bond acceptors and therefore closer to the closure's well-specified regime, are the seed-robust stratum in which the penalty reverses sign ($\Delta\mathrm{MAE} = -0.22 \pm 0.11$ over seeds). We flag this as exploratory rather than a confirmed improvement: across a family of nine simultaneous solvent-class comparisons the amide interval is an uncorrected per-class pair bootstrap, and the three-seed spread alone does not exclude zero; the halogenated class shows a nominal improvement of similar magnitude but is not seed-robust. The robust reading is the *direction* — the asymptotic physics penalty shrinks and can turn negative as solvent hydrogen-bond donation weakens toward the acceptor-dominated regime the closure was built for — not the significance of any single class. This solvent-resolved pattern is the chemistry-resolved counterpart of the same synthetic sweep we used to vary closure fidelity in controlled experiments.

[3] Permutation null (learned coupling destroyed) $-0.90$ to $-0.93$; model-factor $\mathrm{corr}(\hat{\Phi}, \ln \gamma_2) = -0.19$; the Joback reference predicts $T_m > 1000\,\mathrm{K}$ for a substantial fraction of these polycyclics.

[4] The per-seed $\Delta$MAE spans $+0.05$ to $+0.25$, so the point estimate is a loose 3-seed mean, not a tight one; being computed on unrounded per-seed means, the rounded 1.70/1.85 differ by 0.15 rather than 0.14.

Table 9: Per-solvent-class asymptotic physics penalty $\Delta$MAE = MAE(physics) − MAE(control) on the scaffold split (mean ± sd over 3 seeds; positive means the closure adds cost).

| Solvent class | $\Delta$MAE (mean ± sd) |
|---|---|
| Water | $+0.52 \pm 0.35$ |
| Carboxylic acid | $+0.39 \pm 0.16$ |
| Sulfoxide | $+0.26 \pm 0.24$ |
| Nitrile | $+0.21 \pm 0.19$ |
| Alcohol | $+0.15 \pm 0.14$ |
| Aromatic | $+0.03 \pm 0.63$ |
| Hydrocarbon | $+0.02 \pm 0.50$ |
| Halogenated | $-0.29 \pm 0.36$ |
| Amide | $-0.22 \pm 0.11$ |

Table 10: Per-solute-class and novelty-stratum asymptotic physics penalty $\Delta$MAE = MAE(physics) − MAE(control) on the scaffold split (positive means the closure adds cost). Solute-class entries are point estimates; the Tanimoto strata carry the 3-seed spread.

| Solute stratum | $\Delta$MAE |
|---|---|
| Oxygenated | $+0.32$ |
| Polyaromatic | $+0.27$ |
| Heterocycle | $+0.17$ |
| Halogenated-aromatic | $\approx 0$ |
| Charged / salt | $-0.04 \pm 0.19$ |
| Tanimoto $\leq 0.3$ (novel) | $-0.04 \pm 0.18$ |
| Tanimoto $> 0.8$ (familiar) | $+0.51 \pm 0.09$ |

The solute axis shows a compatible but weaker pattern (Table 10). The closure costs most on ordinary neutral organics—oxygenated solutes, polyaromatics, and heterocycles—while an earlier hypothesis, that it should hurt most on charged, high-$|\ln \gamma_2|$ solutes such as salts, did not replicate once the split was corrected: the charged/salt stratum is neutral. The clean penalty falls on neutral organics, not on the ionic tail we had expected to be hardest.

The most suggestive pattern is a novelty inversion that runs opposite to the usual coverage intuition: the physics arm is neutral on the most novel chemistry (nearest-train Tanimoto $\leq 0.3$) and hurts most on the most familiar ($> 0.8$; Table 10). We flag this as exploratory, not a confirmed effect: the seed spread is not a within-bin sampling interval over which solutes fall in the stratum, and the Tanimoto binning is one of several strata families we examine (solvent class, solute class, similarity) without multiplicity control. Read as a direction rather than a tested quantity, it suggests the ceiling on the physics path is not coverage or extrapolation but the closure's systematic misfit—masked on novel solutes by the large errors both arms already incur there, and exposed on easy, well-covered solutes where the direct control is accurate enough for a structural bias in the closure to become the dominant residual—consistent with the same closure-misspecification reading as the rest of the paper. A multiplicity-controlled, within-bin-bootstrapped test is left to future work. Read together, the maps indicate that the physics decomposition incurs cost where the closure is misspecified relative to the solvent chemistry, and that it can only become a net benefit in the narrow acceptor-dominated regime the closure was built to describe.

Table 11: Data-efficiency sweep (train subsampled by solute, fixed scaffold test, seed 42). $\Delta$MAE = MAE(physics) − MAE(direct).

| train fraction | MAE | | $\Delta$MAE | $R^2$ | |
|---|---|---|---|---|---|
| (by solute) | physics | direct | (phys−dir) | physics | direct |
| 0.05 | 2.28 | 2.20 | +0.08 | −0.09 | +0.07 |
| 0.10 | 2.44 | 2.07 | +0.37 | −0.25 | +0.15 |
| 0.25 | 2.93 | 1.81 | +1.12 | −0.75 | +0.38 |
| 0.50 | 3.12 | 1.98 | +1.14 | −0.98 | +0.22 |
| 1.00 | 2.98 | 1.97 | +1.01 | −0.79 | +0.22 |

## G.5 Grounding does not pay off in the low-data regime either

The strongest a-priori case for a physics prior is data scarcity: with few training molecules, an inductive bias toward the correct factorization should beat a data-hungry black box. We test it directly, subsampling the training set *by solute* at fractions 0.05–1.0 and training both the physics-grounded model and DirectGNN on the identical subsample, evaluated on the fixed scaffold test (Table 11, seed 42). The physics arm is *worse at every fraction*, and against the hypothesis the gap is *smallest* at the least data ($\Delta$MAE = +0.08 at 5%) and *grows* with more (+1.0 at full); its $R^2$ is negative throughout. We do not draw strong conclusions from a single-seed sweep with non-monotone physics-arm MAEs, and the light per-fraction budget leaves the physics arm more under-converged than the control, so we take this as *consistent with*, not a sharp confirmation of, the sign rule (Prop. 1). The one inverting metric is top-1 solvent ranking (0.42 vs 0.23 at 5%; the ranking-versus-accuracy split of §G.1), which does not translate into lower error. This is the last regime a physics prior should plausibly help; that it does not is consistent with the broad accuracy null, alongside the structural account—a non-identifiable split through a misspecified closure—as the operative explanation.

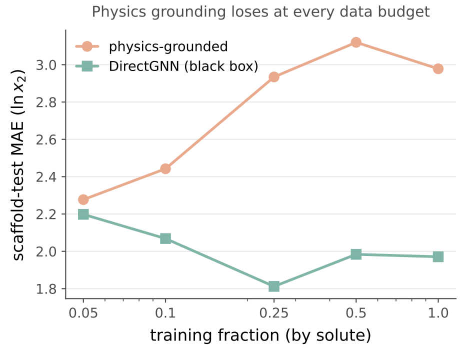


Figure 8: Data-efficiency curve: physics-grounded arm (salmon) and DirectGNN (teal) as a function of training-data fraction.

# H The pKa closure: full construction, measured sweep, and trained comparison

This appendix gives the full second-closure instantiation summarized in §5.7: a fixed Hammett linear-free-energy relationship (LFER) $g(\sigma) = \mathrm{p}K_{a,0} - \rho\,\sigma$ over the Hammett substituent constant $\sigma$ (with $\mathrm{p}K_{a,0}, \rho$ the per-scaffold Hansch–Leo–Taft reaction-series constants), for which Hansch–Leo tabulated constants provide an external oracle $z^\star$. The analogy is close: as with the $\sigma$-profile through COSMO-SAC, a learnable physical descriptor is pushed through a fixed, approximate, differentiable map. The two classic LFER failure modes fall cleanly onto the two error channels of Lemma 2: *resonance saturation*—a curvature in $\sigma$ that the linear closure structurally cannot represent—is a mis-map of a function of $z^\star$ and lands in $B_{\text{closure}}$; *ortho/steric* field effects, absent from the tabulated constants, are information the intermediate lacks and land in $B_{\text{insuff}}$. The chemical scope is a fidelity sweep: meta/para monosubstituted benzoic acids, phenols, and anilinium ions are the near-exact ($B_{\text{closure}}\!\approx\!0$) regime, ortho (insufficiency-dominated) and polyfunctional the low-fidelity one.

This decomposition is directly visible before any training. Pushed through the fixed Hammett closure, *p*-nitrophenol reads $\mathrm{p}K_a \approx 8.3$ against an experimental $\approx 7.1$ (and *p*-nitroaniline overshoots similarly): the linear closure is off by $\sim 1$ unit exactly where through-conjugation ($\sigma^-$) breaks it—$B_{\text{closure}}$ observed in practice.

A controlled semi-synthetic construction on real tabulated $\sigma$ (known ground-truth channels) validates the estimator and reproduces the paper's central distinction in this second closure (Fig. 9). Sweeping closure fidelity with a sufficient intermediate, the grounding gap $\Gamma$ tracks $B_{\text{closure}}$ (the plug-in estimate recovers the truth to within a few percent, e.g. 0.20 vs 0.21, 0.81 vs 0.81), the case where Theorem 1's proxy is valid. Sweeping the ortho/steric channel with a *well-specified* closure gives $B_{\text{closure}} = 0$ throughout yet $\Gamma > 0$ and growing—the pKa realization of the conflation that makes $\Gamma$ an indicator rather than a certificate. The oracle-$\sigma$ pipeline (scaffold-resolved substituent$\to\sigma$ assignment) and this validation extend directly to real data. On the OPERA experimental $\mathrm{p}K_a$ compilation[44] (7,904 records under the same net-neutral filter), applying the same decomposition to the fixed Hammett closure yields a *measured* fidelity sweep: on the clean meta/para series ($n$=158) the closure predicts experimental $\mathrm{p}K_a$ to RMSE 0.55 with the closure at the input-insufficiency floor: the leakage-immune lower bound MSE $-\ B^{\text{up}}_{\text{insuff}}$ is $-0.2$ (95% CI $[-0.4, -0.0]$), i.e. below zero, as a lower bound on the non-negative $B_{\text{closure}}$ must sit when the LFER predicts as well as a flexible within-scaffold regressor—$B_{\text{closure}}$ is not separately resolvable and is consistent with a well-specified closure (per-scaffold systematic offsets $\leq 0.18$). The ortho/heteroaromatic regime ($n$=846) degrades to RMSE 2.26, dominated by input insufficiency. Here the pKa latent is a *scalar* Hammett constant, so—unlike the solubility regime—$B_{\text{insuff}}$ is *point*-identified rather than one-sidedly bounded (§3.2): a within-scaffold regression estimator gives $B_{\text{insuff}} = 4.5$ (95% CI [4.0, 5.1]; $\approx$ 89% of the error—the tabulated $\sigma$ carries no ortho information, exactly the a-priori channel assignment), leaving a small but certified *point* closure misspecification $B_{\text{closure}} = 0.6$ ([0.3, 0.8], a stratified within-scaffold molecule bootstrap). We state $B_{\text{insuff}}$ as a value here, and only here, because it is measured, not bounded (see the resolution note in §3.2). This is the near-exact-to-degraded transition the closure is meant to show, now measured rather than asserted. This places the pKa closure near the phase-diagram boundary in its clean regime.

**Data quality control (uniform, closure-independent).** The trained runs use the canonical full OPERA set (7904 records). A handful of entries are supplied as pre-ionized microstates—an [NH$^+$]-protonated aminopyridine at $\mathrm{p}K_a = -7.78$, a pyrylium [O$^+$], and a few others—inconsistent with a neutral-scaffold Hammett assignment, and they degrade the clean pole (pyridinium RMSE $0.63 \to 3.83$). We drop them with one rule, *net formal charge* $\neq 0$, applied uniformly to both strata. It is closure-independent (it never references the oracle residual, so it cannot be circular) and keeps net-neutral substituents such as nitro [N$^+$][O$^-$]=O. This removes 5 of 163 meta/para rows (restoring the clean pole to RMSE $\approx$0.55) and 352 of 1198 ortho rows (leaving 846; the ortho RMSE is essentially unchanged, so the filter does not manufacture the sign), for 1004 Hammett-amenable rows.

**Certification of the flip.** We train the physics arm (encoder $\to \hat{\sigma} \to$ fixed LFER) on $K$=20 stratified 80/20-

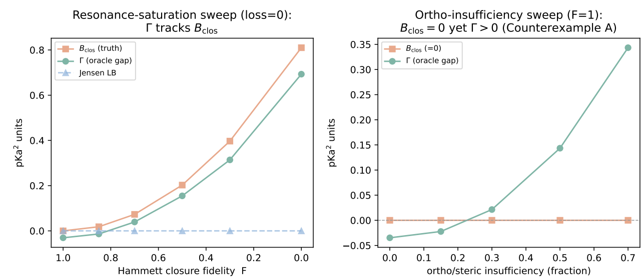


Figure 9: pKa through a Hammett LFER, on a controlled construction with known ground-truth channels. *Left:* closure fidelity varied with a sufficient intermediate; grounding gap $\Gamma$, closure bias $B_{\text{closure}}$, and the assumption-light Jensen bound. *Right:* an ortho/steric insufficiency channel varied with a *well-specified* closure; $B_{\text{closure}} = 0$ throughout while $\Gamma > 0$ and grows.

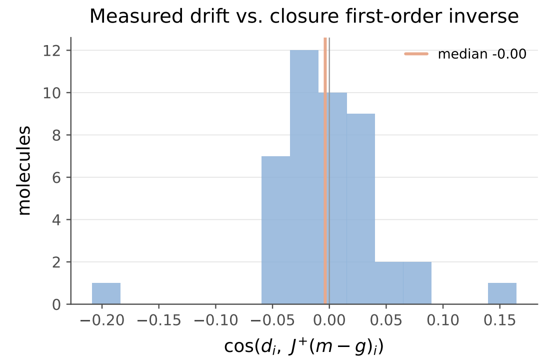


Figure 10: Per-molecule cosine between the measured drift $\hat{\sigma} - \sigma$ and the first-order closure compensation $J^{+}(m - g)$; the distribution is centred on zero (median $-0.00$).

within-scaffold splits—a Bemis–Murcko split collapses the meta/para pole to two ring systems and is unstable—and read the paired per-molecule error against the deterministic oracle, per stratum. Meta/para: learned MAE $0.48 \pm 0.09$ vs oracle $0.315 \pm 0.07$, gap $-0.165$, oracle better in 19/20 splits (scaffold-cluster and split bootstraps both exclude zero, $P(\text{oracle worse}) < 0.01$). Ortho: learned $0.83 \pm 0.07$ vs oracle $1.62 \pm 0.08$, gap $+0.79$, oracle worse in 20/20 splits ($P(\text{oracle worse}) = 1.00$, 90% CI $[+0.57, +0.95]$).

**The crossover law and its scope boundary.** A competence sweep (train fraction $\in \{1.0, 0.5, 0.25, 0.12\}$ and, as the low-competence extreme, scaffold extrapolation; six seeds each) confirms the sign is set by the crossover of the learned error against the fidelity-fixed oracle threshold. On meta/para the learned arm rises from 0.46 to 0.98 as data shrinks but never reaches the 0.31 oracle threshold, so grounding helps throughout. On ortho the learned arm stays below the 1.62 threshold across the whole interpolation range (0.85 at full data, 1.31 at 12%), so grounding hurts throughout; only scaffold extrapolation, where $\hat{\sigma}$ breaks to $2.6 \pm 2.0$, pushes it back above the threshold and reverts the sign. The flip is therefore a within-distribution, fidelity-driven crossover, and the extrapolation reversion is predicted by the same law rather than a contradiction of it. A trained physics-vs-DirectGNN comparison points the same way overall but has weak absolute errors and an untuned control, so we treat it only as a complement; the load-bearing second-domain results are the decomposition above and the certified flip.

## H.1 Diagnostics that do not work, and why

**The drift is not the closure's signed first-order inverse—but the test is under-resolved.** One might hope the compensating drift is the closure's linear-response inverse, $\hat{\sigma} - \sigma \approx J^{+}(m - g(\sigma))$ (with $J = \partial g/\partial\sigma$), predictable a priori from $J$ before any solubility training. On the matched set the per-molecule drift shows no signed alignment with $J^{+}(m - g)$ (median cosine $-0.00$; Fig. 10).[5] We read this as a signed negative only, and do not build on it, for two reasons the test cannot escape. First, the matched set is sparse (median one pair per molecule), so each per-molecule Jacobian subspace is one-dimensional and the dimension-matched row-space controls are uninformative—a resolved direction common across molecules makes a molecule's "own" and "other" subspaces coincide. Second, and more fundamentally, the single-pair $J^{+}(m - g)$ is a poor *estimator* of the first-order compensation: the activity Fisher is near rank-one (PR $\approx 1.4$ of 51, §5.4), so the closure's first-order inverse is ill-posed along all but one direction. A null here is therefore uninformative about whether the drift is first-order, not evidence that it is not. The load-bearing claim is the under-determination of $\sigma$ by activity (§5.4), not the geometry of the drift.

The instrument behind §5.4—the rank of the activity Fisher $J^{\top}J$ ($J = \partial \ln \gamma_2^{\infty}/\partial\sigma$), read as a participation ratio $\mathrm{PR} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$—is cheap and oracle-free, which is its appeal for probing any physics-grounded latent. It is also easy to misread in three ways that inflate a null or a rank in the flattering direction, silently; we record them because the corrections are not obvious.

**(i) Finite differences inflate the participation ratio.** PR is a *tail* statistic, governed by the small eigenvalues. A closure evaluated by an unrolled fixed point carries a small, step-*independent* gradient noise (from the convergence tolerance, not truncation), so central

[5] The residual COSMO-SAC closure runs a segment-activity fixed point, so its autograd Jacobian is numerically unstable ($\|\partial g/\partial\sigma_{\text{solvent}}\| \sim 10^{15}$ at infinite dilution); $J$ is computed by finite differences on the converged closure.

finite differences lift the exactly- and near-zero eigenvalues onto a floor and PR rises. On the differentiable COSMO-SAC-2002 layer, autograd and finite differences on *identical* profiles give PR = 1.00 and 1.25—a 25% inflation from noise alone, though the per-gradient cosine is 0.995 (the direction is right, the tail is not). It is worse on a typed $(3 \times 51)$ profile basis: a closure that ignores the typing (2002) has 102 exactly-redundant coordinates that finite differences decorrelate into a spurious floor, inflating its PR from 1.0 toward 2.0, while a closure that uses the typing (2010) has no such redundancy—so the same step inflates the two *asymmetrically* and can manufacture a rank ordering with no physical content. *Use autograd where the closure is differentiable; with only finite differences, read* PR *against a known-rank or permuted control, never absolutely.*

**(ii) "Own vs. other" controls are powerless when the resolved direction is common.** A dimension- and smoothness-matched control aligns a molecule's drift against its own predicted compensation and against other molecules'. It has power only if the predictions differ. But when the aggregate Fisher is rank-one (§5.4) the resolved direction is *common*, so "own" and "other" point almost the same way and the diagonal must sit inside the null—$p \approx 0.5$ regardless of the truth. On the matched set sparsity compounds this: a median of one pair per molecule makes each per-molecule Jacobian subspace one-dimensional, so "own" and "other" subspaces coincide exactly. *Verify a control has power before reading its null: if the compared predictions have median cosine near one, the test cannot reject.*

**(iii) Energy-in-subspace baselines need the effective rank and a cluster null.** A random zero-mean vector in $D$ dimensions places $k_{\text{eff}}/(D-1)$ of its energy in a subspace of *effective* rank $k_{\text{eff}}$ (the participation ratio of its singular values), which for near-collinear Jacobian rows is far below the nominal row count. On the matched set the per-molecule subspace is effectively one-dimensional, so the baseline is ≈2% and the observed 5.5% is a two-to-threefold *enrichment*, not the "no concentration" a raw own-versus-other equality suggests: own equals other because both one-dimensional subspaces are the same common direction, not because the drift avoids the resolved one. And with that direction common, $N$ enriched molecules are not $N$ independent draws—the enrichment needs a cluster null (§5.2), without which its significance is over-stated by $\sqrt{N/N_{\text{eff}}}$. We therefore report the 5.5% as evidence in neither direction. *Baseline by effective rank; cluster by the shared direction.*

Each failure turns a soft spectral quantity into a hard structural claim—the mode the surviving statements of §5.4 are written to avoid, by carrying their null model and a power check inline.

# I Data provenance and reproducibility

**Sources.** Solubility: BigSolDB 2.0[29]. Crystal $T_m/\Delta H_{\text{fus}}$: an open melting-point pool plus a group-contribution prior. True $\sigma$-profiles: the VT-2005 database (ingested to `sigma_profiles.csv`, 1432 compounds, with the COSMO cavity volume $V_{\text{cosmo}}$ for the combinatorial term). These are quantum-chemically computed for a single reference conformer and protonation state; any mismatch to the experimental measurement conditions is noise in $z^\star$ that inflates $B_{\text{insuff}}$ (an imperfect intermediate) rather than $B_{\text{closure}}$, so it is *conservative* for the closure-dominance verdict. Infinite-dilution activity coefficients: ThermoML. We checked whether enlarging the reference $\sigma$-profile set would grow the matched activity corner: of the 298 K ThermoML IDAC pairs, only two solutes lack a VT-2005 profile while their solvent has one, so computing further reference profiles does not materially enlarge the matched set—the binding constraint is joint IDAC×profile coverage, not profile availability. Second-domain pKa: the OPERA experimental compilation[44] (`pKa_QR.sdf`, 7904 QSAR-ready records, public-domain US EPA; cited, not redistributed); the uniform net-formal-charge filter reduces this to the 1004 Hammett-amenable rows (158 meta/para, 846 ortho/heteroaromatic) analysed in §H. The Bemis–Murcko scaffold split follows the main text; because the seeded split is not stable across pipeline versions (§3), every number in this paper is computed on a *single* committed split, whose molecule-level hash is recorded with the released artifacts.[6] Melting-point tables are auto-detected as °C or K to avoid a +273 K double-conversion.

**Compute ledger.** The headline $\sigma$-grounding comparison (Fig. 2, Table 4) is a full-corpus GPU run (three seeds); the closure/insufficiency decomposition (§5.2), the Fisher audit (§G.2), the noise floor, the encoder probe, and the temperature and mechanism diagnostics of §F are CPU-reproducible from committed artifacts. Each figure and table names the script that regenerates it from `results/`; large prediction CSVs and model checkpoints are deposited separately ([Zenodo DOI]) rather than in the repository.

**Seeds and determinism.** Controlled comparisons use seeds {42, 43, 44} and are reported on a cross-arm intersection lock (rows supervised and finite in every arm); the surrogate isolation (§5.4) uses seeds {0, 1, 2}. The

[6] The three-seed surrogate isolation (§5.4) was produced on a separate run whose per-run split provenance was not logged; its matched set is the same $n$=44 molecules, derivable from the committed split independently of the model, so it is consistent with the reported split. We record the split hash in the runner so future runs are auditable directly.

decomposition is seed-independent (its closure has no learnable parameters). One artifact caveat: a save-time bug truncated the released per-row prediction file for the seed-44 oracle solubility arm to 295 rows, a small fraction of the test set. The three-seed headline summaries (Fig. 2, Table 4) were aggregated at run time, before that file was written, so they are unaffected; only the one on-disk artifact is partial, and any per-row re-analysis of the solubility arms therefore uses the two complete seeds (42, 43). The learned-$\sigma$ and DirectGNN solubility files and the activity-level ($n$=60, $n$=44) artifacts are complete for every seed.

# References


[1] Yue Yang and Shiang-Tai Lin. Physics-embedded machine learning model for phase equilibrium prediction in multicomponent systems. *Journal of Chemical Information and Modeling*, 2025. doi: 10.1021/acs.jcim.5c01804.

[2] Florence H. Vermeire, Yunsie Chung, and William H. Green. Predicting solubility limits of organic solutes for a wide range of solvents and temperatures. *Journal of the American Chemical Society*, 144(24):10785–10797, 2022. doi: 10.1021/jacs.2c01768.

[3] Lucas Attia, Jackson W. Burns, Patrick S. Doyle, and William H. Green. Data-driven organic solubility prediction at the limit of aleatoric uncertainty. *Nature Communications*, 16:7497, 2025. doi: 10.1038/s41467-025-62717-7. URL `https://doi.org/10.1038/s41467-025-62717-7`. FAST-SOLV model; earlier ChemRxiv 2024 preprint doi:10.26434/chemrxiv-2024-93qp3.

[4] Benedikt Winter, Clemens Winter, Johannes Schilling, and André Bardow. A smile is all you need: Predicting limiting activity coefficients from SMILES with natural language processing. *Digital Discovery*, 1(6):859–869, 2022. doi: 10.1039/D2DD00058J.

[5] Edgar Ivan Sanchez Medina, Steffen Linke, Martin Stoll, and Kai Sundmacher. Gibbs–helmholtz graph neural network: capturing the temperature dependency of activity coefficients at infinite dilution. *Digital Discovery*, 2(3):781–798, 2023. doi: 10.1039/D2DD00142J. URL `https://doi.org/10.1039/D2DD00142J`.

[6] George Em Karniadakis, Ioannis G. Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3:422–440, 2021. doi: 10.1038/s42254-021-00314-5.

[7] Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019. doi: 10.1016/j.jcp.2018.10.045.

[8] Jared Willard, Xiaowei Jia, Shaoming Xu, Michael Steinbach, and Vipin Kumar. Integrating scientific knowledge with machine learning for engineering and environmental systems. *ACM Computing Surveys*, 55(4):1–37, 2022. doi: 10.1145/3514228.

[9] Lei Ma, Nicolas Boullé, Yu-Sen Yang, Hao Wu, and Ling Guo. Physics-guided correction for operator learning under model misspecification. *arXiv preprint arXiv:2606.03469*, 2026. doi: 10.48550/arXiv.2606.03469.

[10] Aditi S. Krishnapriyan, Amir Gholami, Shandian Zhe, Robert M. Kirby, and Michael W. Mahoney. Characterizing possible failure modes in physics-informed neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021.

[11] Arvind Mohan, Ashesh Chattopadhyay, and Jonah Miller. What you see is not what you get: Neural partial differential equations and the illusion of learning. *arXiv preprint arXiv:2411.15101*, 2024. doi: 10.48550/arXiv.2411.15101.

[12] Anita Mahinpei, Justin Clark, Isaac Lage, Finale Doshi-Velez, and Weiwei Pan. Promises and pitfalls of black-box concept learning models. *arXiv preprint arXiv:2106.13314*, 2021.

[13] Andrei Margeloiu, Matthew Ashman, Umang Bhatt, Yanzhi Chen, Mateja Jamnik, and Adrian Weller. Do concept bottleneck models learn as intended? *arXiv preprint arXiv:2105.04289*, 2021.

[14] Marton Havasi, Sonali Parbhoo, and Finale Doshi-Velez. Addressing leakage in concept bottleneck models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022.

[15] Enrico Parisini, Tapabrata Chakraborti, Chris Harbron, Ben D. MacArthur, and Christopher R. S. Banerji. Leakage and interpretability in concept-based models. *arXiv preprint arXiv:2504.14094*, 2025. doi: 10.48550/arXiv.2504.14094.

[16] Mateo Espinosa Zarlenga, Gabriele Dominici, Pietro Barbiero, Zohreh Shams, and Mateja Jamnik. Avoiding leakage poisoning: Concept interventions under distribution shifts. *arXiv preprint arXiv:2504.17921*, 2025. doi: 10.48550/arXiv.2504.17921.

[17] Marc C. Kennedy and Anthony O'Hagan. Bayesian calibration of computer models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 63(3):425–464, 2001. doi: 10.1111/1467-9868.00294.

[18] Jenny Brynjarsdóttir and Anthony O'Hagan. Learning about physical parameters: the importance of model discrepancy. *Inverse Problems*, 30 (11):114007, 2014. doi: 10.1088/0266-5611/30/11/114007.

[19] Allan H. Murphy. A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600, 1973. doi: 10.1175/1520-0450(1973)012<0595:ANVPOT>2.0.CO;2.

[20] Morris H. DeGroot and Stephen E. Fienberg. The comparison and evaluation of forecasters. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 32(1–2):12–22, 1983. doi: 10.2307/2987588.

[21] Halbert White. Maximum likelihood estimation of misspecified models. *Econometrica*, 50(1):1–25, 1982. doi: 10.2307/1912526.

[22] Mark K. Transtrum, Benjamin B. Machta, Kevin S. Brown, Bryan C. Daniels, Christopher R. Myers, and James P. Sethna. Perspective: Sloppiness and emergent theories in physics, biology, and beyond. *The Journal of Chemical Physics*, 143(1):010901, 2015. doi: 10.1063/1.4923066.

[23] Ryan N. Gutenkunst, Joshua J. Waterfall, Fergal P. Casey, Kevin S. Brown, Christopher R. Myers, and James P. Sethna. Universally sloppy parameter sensitivities in systems biology models. *PLoS Computational Biology*, 3(10):e189, 2007. doi: 10.1371/journal.pcbi.0030189.

[24] A. Raue, C. Kreutz, T. Maiwald, J. Bachmann, M. Schilling, U. Klingmüller, and J. Timmer. Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics*, 25(15):1923–1929, 2009. doi: 10.1093/bioinformatics/btp358.

[25] Alejandro F. Villaverde, Antonio Barreiro, and Antonis Papachristodoulou. Structural identifiability of dynamic systems biology models. *PLoS Computational Biology*, 12(10):e1005153, 2016. ISSN 1553-7358. doi: 10.1371/journal.pcbi.1005153.

[26] Alexander D'Amour, Katherine Heller, Dan Moldovan, Ben Adlam, Babak Alipanahi, Alex Beutel, Christina Chen, Jonathan Deaton, Jacob Eisenstein, Matthew D. Hoffman, Farhad Hormozdiari, Neil Houlsby, Shaobo Hou, Ghassen Jerfel, Alan Karthikesalingam, Mario Lucic, Yian Ma, Cory McLean, Diana Mincu, Akinori Mitani, Andrea Montanari, Zachary Nado, Vivek Natarajan, Christopher Nielson, Thomas F. Osborne, Rajiv Raman, Kim Ramasamy, Rory Sayres, Jessica Schrouff, Martin Seneviratne, Shannon Sequeira, Harini Suresh, Victor Veitch, Max Vladymyrov, Xuezhi Wang, Kellie Webster, Steve Yadlowsky, Taedong Yun, Xiaohua Zhai, and D. Sculley. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23(226):1–61, 2022. URL `https://www.jmlr.org/papers/v23/20-1335.html`.

[27] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. ISSN 2666-3899. doi: 10.1016/j.patter.2023.100804.

[28] Divyat Mahajan, Mohammad Pezeshki, Charles Arnal, Ioannis Mitliagkas, Kartik Ahuja, and Pascal Vincent. Compositional risk minimization. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 42503–42553. PMLR, 2025.

[29] Lev Krasnov, Dmitry Malikov, Marina Kiseleva, Sergei Tatarin, Sergey Sosnin, and Stanislav Bezzubov. BigSolDB 2.0, dataset of solubility values for organic compounds in different solvents at various temperatures. *Scientific Data*, 12:1236, 2025. doi: 10.1038/s41597-025-05559-8.

[30] Shiang-Tai Lin and Stanley I. Sandler. A priori phase equilibrium prediction from a segment contribution solvation model. *Industrial & Engineering Chemistry Research*, 41(5):899–913, 2002. doi: 10.1021/ie001047w.

[31] Eric Mullins, Richard Oldland, Y. A. Liu, Shu Wang, Stanley I. Sandler, Chau-Chyun Chen, Michael Zwolak, and Keith C. Seavey. Sigma-profile database for using COSMO-based thermodynamic methods. *Industrial & Engineering Chemistry Research*, 45(12):4389–4415, 2006. doi: 10.1021/ie060370h.

[32] Michael Frenkel, Robert D. Chirico, Vladimir V. Diky, Xinjian Yan, Qian Dong, and Chris Muzny. ThermoML: An XML-based approach for storage and exchange of experimental and critically evaluated thermophysical and thermochemical property data. *Journal of Chemical Information and Computer Sciences*, 43(6):1344–1359, 2003.

[33] Heinz W. Engl, Martin Hanke, and Andreas Neubauer. *Regularization of Inverse Problems*, volume 375 of *Mathematics and Its Applications*. Kluwer Academic Publishers, 1996.

[34] Per Christian Hansen. *Rank-Deficient and Discrete Ill-Posed Problems: Numerical Aspects of Linear Inversion.* SIAM Monographs on Mathematical Modeling and Computation. SIAM, 1998.

[35] Johannes Schwab, Stephan Antholzer, and Markus Haltmeier. Deep null space learning for inverse problems: Convergence analysis and rates. *Inverse Problems*, 35(2):025008, 2019. doi: 10.1088/1361-6420/aaf14a.

[36] Henri Renon and J. M. Prausnitz. Local compositions in thermodynamic excess functions for liquid mixtures. *AIChE Journal*, 14(1):135–144, 1968. doi: 10.1002/aic.690140124.

[37] Aage Fredenslund, Russell L. Jones, and J. M. Prausnitz. Group-contribution estimation of activity coefficients in nonideal liquid mixtures. *AIChE Journal*, 21(6):1086–1099, 1975. doi: 10.1002/aic.690210607.

[38] Ian H. Bell, Erik Mickoleit, Chieh-Ming Hsieh, Shiang-Tai Lin, Jadran Vrabec, Cornelia Breitkopf, and Andreas Jäger. A benchmark open-source implementation of COSMO-SAC. *Journal of Chemical Theory and Computation*, 16(4):2635–2646, 2020. doi: 10.1021/acs.jctc.9b01016.

[39] Chieh-Ming Hsieh, Stanley I. Sandler, and Shiang-Tai Lin. Improvements of COSMO-SAC for vapor–liquid and liquid–liquid equilibrium predictions. *Fluid Phase Equilibria*, 297(1):90–97, 2010. doi: 10.1016/j.fluid.2010.06.011.

[40] Chieh-Ming Hsieh, Shiang-Tai Lin, and Jadran Vrabec. Considering the dispersive interactions in the COSMO-SAC model for more accurate predictions of fluid phase behavior. *Fluid Phase Equilibria*, 367:109–116, 2014. doi: 10.1016/j.fluid.2014.01.032.

[41] Robin Fingerhut, Wei-Lin Chen, Andre Schedemann, Wilfried Cordes, Jürgen Rarey, Chieh-Ming Hsieh, Jadran Vrabec, and Shiang-Tai Lin. Comprehensive assessment of COSMO-SAC models for predictions of fluid-phase equilibria. *Industrial & Engineering Chemistry Research*, 56(35):9868–9884, 2017. doi: 10.1021/acs.iecr.7b01360.

[42] Naoya Takeishi and Alexandros Kalousis. Physics-integrated variational autoencoders for robust and interpretable generative modeling. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021.

[43] Christian Internò, Jumpei Yamaguchi, Loren Amdahl-Culleton, Markus Olhofer, David Klindt, and Barbara Hammer. The observer effect in world models: Invasive adaptation corrupts latent physics. *arXiv preprint arXiv:2602.12218*, 2026. doi: 10.48550/arXiv.2602.12218.

[44] Kamel Mansouri, Neil F. Cariello, Alexandru Korotcov, Valery Tkachenko, Christopher M. Grulke, Catherine S. Sprankle, David Allen, Warren M. Casey, Nicole C. Kleinstreuer, and Antony J. Williams. Open-source QSAR models for pKa prediction using multiple machine learning approaches. *Journal of Cheminformatics*, 11:60, 2019. doi: 10.1186/s13321-019-0384-1.

[45] Keisuke Hirano, Guido W. Imbens, and Geert Ridder. Efficient estimation of average treatment effects using the estimated propensity score. *Econometrica*, 71(4):1161–1189, 2003. doi: 10.1111/1468-0262.00442.

[46] James M. Robins and Ya'acov Ritov. Toward a curse of dimensionality appropriate (CODA) asymptotic theory for semi-parametric models. *Statistics in Medicine*, 16(3):285–319, 1997.

[47] Ladislav Rampášek, Mikhail Galkin, Vijay Prakash Dwivedi, Anh Tuan Luu, Guy Wolf, and Dominique Beaini. Recipe for a general, powerful, scalable graph transformer. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022.

[48] A. W. van der Vaart. *Asymptotic Statistics*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 1998. Chapter 25 (semiparametric efficiency: efficient score and information).